

Nucleation-dependent propagation of Polycomb modifications emerges during the *Drosophila* maternal to zygotic transition

Natalie Gonzaga-Saavedra, Eleanor A Degen, Isabella V Soluri, Corinne Croslyn, Shelby A Blythe 

Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, United States • Department of Molecular Biosciences, Northwestern University, Evanston, United States • National Institute for Theory and Mathematics in Biology, Northwestern University and the University of Chicago, Chicago, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study employed state-of-the-art quantitative imaging and genomics approaches to address a **fundamental** question regarding the establishment of Polycomb domains during *Drosophila* embryogenesis. The critical developmental stage was pinpointed to the maternal-to-zygotic transition, rather than earlier stages, providing clarification for the field. The roles of two factors, Zelda and GAGA-factor, were investigated, which reveal that Zelda, but not GAGA-factor, contributes to this process. These **compelling** findings have implications for chromatin and developmental biology.

<https://doi.org/10.7554/eLife.108371.1.sa9>

Abstract

During zygotic genome activation (ZGA) in *Drosophila*, broad domains of Polycomb-modified chromatin are rapidly established across the genome. Here, we investigate the spatial and temporal dynamics by which Polycomb group (PcG) histone modifications, H3K27me3 and H2Aub, emerge during early embryogenesis. Using ChIP-seq and live imaging of CRISPR-engineered GFP-tagged PcG components, we show that PRC2-dependent H3K27me3 accumulates adjacent to a subset of E(z)-bound prospective Polycomb Response Elements (PREs) beginning in nuclear cycle 14 (NC14), with patterns indicative of nucleation followed by spreading. Surprisingly, PRE-binding factors Pho, Combgap, and GAGA-factor are excluded from interphase nuclei prior to NC10 despite nuclear localization of E(z) throughout early interphases. Loss-of-function studies further demonstrate that GAGA-factor is largely dispensable for PcG domain establishment, whereas the pioneer factor Zelda is required for proper deposition of H3K27me3 and H2Aub at a subset of Polycomb domains. The role of Zelda at Polycomb domains is context-dependent; a large subset of targets requires Zelda not for PcG factor recruitment, but instead to license a loaded PRE to deposit H3K27me3 and

H2Aub. Our findings support a model where licensing of PcG domains is an initial step in the regulatory processes governing Polycomb-regulated developmental genes.

Introduction

During the maternal to zygotic transition, the genomes of two highly specialized cells, sperm and egg, are epigenetically reprogrammed in order to support the development of the zygote. Following an initial period of post-fertilization mitotic proliferation, embryos undergo genome activation and initiate zygotic gene regulatory programs against the background of the reprogrammed epigenetic landscape. Many of the genes involved in cell fate specification and pattern formation become associated with chromatin modifications deposited by the Polycomb group (PcG), whose function is ultimately necessary for maintaining these genes in a heritable, transcriptionally silenced state. While the mechanism for PcG function for maintenance of silenced states has been intensely studied, the rules whereby genomic loci are selected for Polycomb silencing, and the mechanisms for establishing these genomic compartments in early embryogenesis remain unclear.

PcG factors associate in multi-subunit complexes that carry out distinct steps required for the maintenance of long-term repression (1,2). Polycomb repressive complexes 1 and 2 (PRC1 and 2) mediate the reading and writing of histone post-translational modifications at the heart of the PcG system (3,4). PRC2 mediates methylation of Histone H3 lysine 27 (H3K27) through the histone methyltransferase subunit Enhancer of Zeste (E(z)) (5,6). Tri-methylation of H3K27 (H3K27me3) deposited by E(z) can be ‘read’ by additional PcG factors to allosterically enhance E(z) catalysis and to promote the spreading of H3K27 methylation across broader genomic regions which can span tens to hundreds of kilobase-pairs (7–9). The ubiquitylation of Histone H2A (H2Aub) by the Sce/dRing1 component of PRC1 can also lead to allosteric enhancement of E(z) catalysis (10,11). Targeting of PRC1 and 2 to sites designated for PcG regulation is driven in part by *cis*-regulatory sequences termed Polycomb Response Elements (PREs) which are bound by several sequence-specific *trans*-acting factors including but not limited to Pleiohomeotic (Pho), Pho-like (Phol), Combgap (Cg), and GAGA-factor (GAF) (12–19). In proliferating cells, functional PRE sequences are necessary for maintaining PcG states across successive cell divisions (20,21), supported by allosteric enhancement from these read-write feedback loops.

Early embryos present several challenges for the establishment and maintenance of PcG modifications. *Drosophila* embryos initially undergo thirteen rapid, metasynchronous nuclear divisions before undergoing cell cycle pause during nuclear cycle 14 (NC14, cellular blastoderm stage) and completing zygotic genome activation (ZGA) (22). Over the cleavage period, embryos gradually establish patterns of chromatin accessibility that will sustain the initial rounds of transcriptional activity that drive developmental patterning (23). These patterns of accessibility are determined in whole or in part by specialized transcription factors termed pioneer factors that operate at distinct timepoints to gradually install the initial chromatin accessibility state (24–27). Genome-wide assessments of H3K27me3 deposition in early *Drosophila* embryos have yielded two distinct models for the establishment of PcG modifications. In one case, H3K27me3 modifications are observed to arise *de novo* only during mid nuclear cycle 14 (NC14), which is coincident with the acquisition of substantial chromatin accessibility and cell cycle lengthening that accompanies ZGA (28–30). In another case, patterns of H3K27me3 deposition have been argued to be transmitted to offspring through the maternal germline, with retention of H3K27me3 at some level throughout the cleavage divisions (31,32). We wished to investigate further the discrepancy between these observations and approach the problem from the mechanistic standpoint of existing models for PcG maintenance. In the following, we have addressed the timing of PcG modifications over the course of *Drosophila* ZGA, with a specific focus on the availability of PRE-binding factors, and the relationship of PcG modification deposition to pioneer factors known to drive establishment of the initial zygotic chromatin state.

Results

Rapid Establishment of Polycomb histone modifications at prospective Polycomb Response Elements during ZGA

To investigate the kinetics of establishment of PcG modification of the zygotic genome, we performed ChIP-seq in cellular blastoderm stage (NC14) embryos for PcG-dependent histone modifications (H3K27me₃, -me₁, and H2Aub, **Figure 1A**). To identify loci where deposition of PcG modifications could be expected, we also measured the blastoderm stage enrichment of the PRC2 component E(z), as well as the core PRE component Pho, additional PRE binding factors, Combgap (Cg) and GAGA-Factor (GAF), and the pioneer factor Zelda (**Figure 1A**, Zelda ChIP-seq data from (33)). To measure E(z), Cg, GAF, and Pho binding, we performed ChIP-seq for each individual factor on chromatin derived from embryos expressing a GFP chimera for the corresponding factor generated by CRISPR/Cas9 genome editing (see Materials and Methods). The engineered EGFP-E(z), EGFP-Cg, EGFP-GAF, and Pho-sfGFP chimeric alleles are all viable and fertile as homozygotes. Measurement of these factors identifies a set of 4576 E(z) peaks that represent prospective targets of PcG regulation, only a subset of which have been functionally validated as PREs. As expected, E(z) peaks also often correspond to binding sites for either Pho, GAF and/or Cg (**Figure 1A**). Overall, 86.6% (n = 3966) E(z) peaks overlap with peaks of Pho, Cg, or GAF in different combinations of factors. PcG-modified domains are nucleated by recruitment of PRC1/2 to PREs, resulting in local modification of neighboring nucleosomes and subsequent spreading of modifications across topologically delimited segments of the genome. To estimate the extent of these broader regions of H3K27me₃, we called using a Hidden Markov Model approach (STAN) a set of PcG “domains” from our late NC14 H3K27me₃ dataset as described previously (34,35). In blastoderm stage chromatin we identify 255 broad PcG domains in total with median length of 19.2 kb (range: 4.2 to 350 kb). Of these PcG domains, 93% (n = 237) contain one or more E(z) peaks. Notably, not all E(z) peaks are found in PcG domains: 72.3% (n = 3312) of E(z) peaks do not associate with above-threshold H3K27me₃ ChIP-seq signal by the end of NC14 (**Figure 2**). In general, E(z) peaks outside of PcG domains were found to localize to the transcriptional start sites in promoters of genes decorated with RNA Polymerase 2 phosphorylated at position 5 of the C-terminal domain heptapeptide repeat as well as promoter-proximal Histone H3 Lysine 4 mono-, di-, and tri-methylation and Histone H3K27 acetylation (**Figure 2** and Supplemental Figure 1). On average, E(z) and Pho binding within PcG domains yields higher ChIP-seq enrichment, and these show evidence of long-distance “spreading” (**Figure 2B**). Scoring occurrences of DNA binding factors at E(z) peaks within PcG domains, only Pho and Zelda demonstrate significant enrichment within PcG domains, whereas Cg and GAF are more enriched at E(z) peaks outside of PcG domains. Based on these observations, the set of E(z) peaks can be classified into at least two categories based either on occurrence within a PcG domain, or outside of the domain. We chose to focus on the characteristics of E(z) peaks within PcG domains.

Consistent with prior reports (28–30), we observe that H3K27me₃ is strongly detected by mid- to late-NC14 adjacent to E(z) peaks and more broadly across PcG domains but is not strongly detected on NC13 and early NC14 chromatin (**Figure 1A–C**). This pattern of accumulation could be due either to limited access of PcG components to chromatin prior to NC14 or could reflect the expected slow kinetic maturation of sub-trimethylated histone species near sites of recruitment for PcG complexes (29,36–38). To distinguish these possibilities, we also performed ChIP-seq against H3K27me₁ and observed a broadly distributed signal surrounding E(z) peaks at NC13 and early NC14 that is gradually depleted from sites that acquire H3K27me₃ by mid- and late-NC14 (**Figure 1B**). At the domain scale, H3K27me₁ abundance is on average higher within domains than in flanking regions during NC13 and early NC14, and this average signal is incrementally reduced at mid- and late-NC14 (**Figure 1C**). The anti-correlation between H3K27me₁ and -me₃ within domains is consistent with a mechanism where the rapid rate of cleavage stage cell cycles

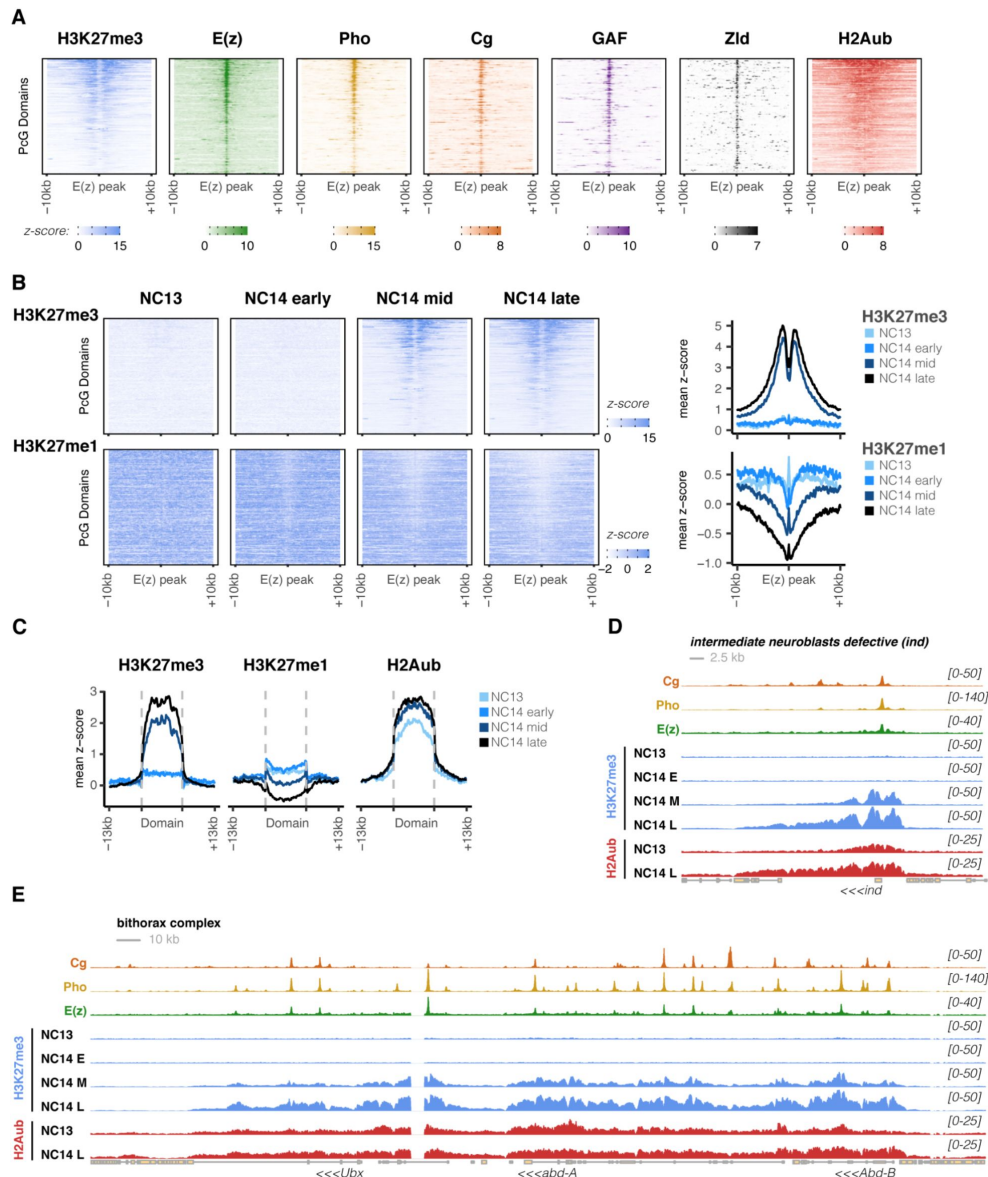


Figure 1

Timing of establishment of PcG modifications at ZGA.

A) Heatmaps are plotted for ChIP-seq measurements in cellular blastoderm staged embryos for the indicated factors (top) over a set of E(z) peaks localized within PcG Domains, flanked by ± 10 kb. One representative E(z) peak per domain was selected for plotting. Heatmap rows are identically ordered from top to bottom by overall average H3K27me3 intensity per peak. E(z), Pho, Cg, and GAF ChIP-seq are all performed using an anti-GFP antibody on CRISPR-engineered embryos expressing GFP-tagged chimeric alleles of the target protein. Zld ChIP-seq was previously reported (33). Data were standardized prior to plotting and colorbars (bottom) indicate binding intensity in units of standard deviations. B) Heatmaps are plotted for standardized ChIP-seq measurements in embryos collected at the indicated stages (top) for either H3K27me3 (top row), or H3K27me1, bottom row. The average standardized ChIP-seq signal per time point is plotted at right. C) The average standardized ChIP-seq signal for timecourse measurements of H3K27me3, -me1, and H2Aub are plotted over entire PcG domains. The domains are first divided into 100 bins and standardized ChIP-seq signal was averaged per bin prior to plotting average signal across domains ± 13 kb flanking sequence. D) Visualization of the PcG domain containing the *ind* locus and counts-per-million (CPM) normalized ChIP-seq measurements for Cg, Pho, E(z) at cellular blastoderm, and timecourse measurements of H3K27me3 (NC13 - NC14 late), and H2Aub (NC13, NC14 late). ChIP target is indicated at left and the display range of the CPM normalized ChIP data is indicated at right. Scale bar: 2.5 kb E) Visualization of the PcG domain containing the bithorax complex for the same CPM-normalized ChIP tracks as in panel D. Scale bar: 10 kb.

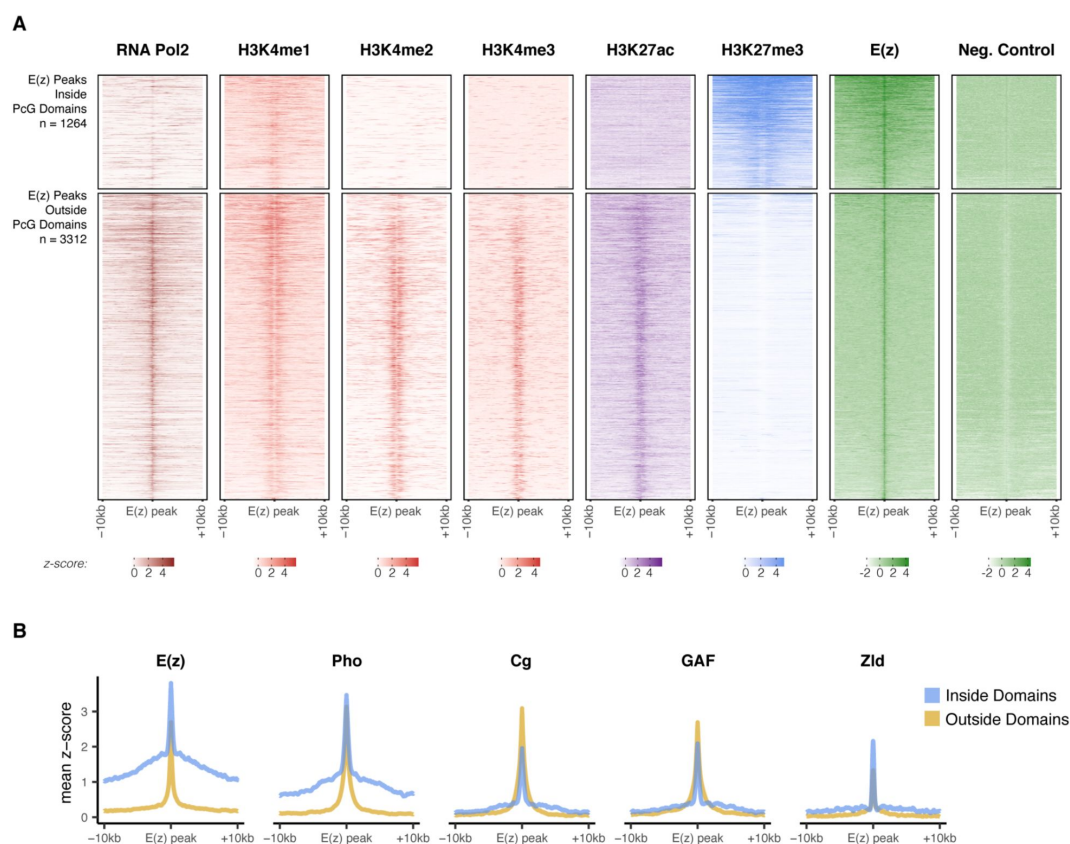


Figure 2

E(z) peaks outside of PcG Domains generally associate with active promoters.

A: Heatmaps for the indicated ChIP targets were plotted centered over all 4576 E(z) peaks (± 10 kb) and subset by membership within a PcG domain. Data were z-scored prior to plotting, and the plotted intensity range is indicated by the colorbar beneath each respective heatmap. Peaks are ordered on the y-axis by the overall intensity of E(z) binding in the region shown. A negative control ChIP (IP for GFP on wild type chromatin) is shown at the far right. While E(z) peaks within domains are enriched for H3K27me3, peaks outside of domains show a high degree of correlation with marks of active transcription including RNA Pol2 (CTD pSer5), H3K4me1/2/3, and H3K27 acetylation. B: The mean enrichments of E(z) and DNA binding factors over E(z) peaks ± 10 kb are plotted for peaks inside (blue) and outside (yellow) of PcG domains.

Factor	Number of Factor Peaks overlapping any E(z) Peak (%)	Number of Factor Peaks overlapping an E(z) peak in a PcG domain (%)	Enrichment in E(z) PcG domains: (log2[Odds Ratio])	Enrichment in E(z) PcG domains: (p-value)
Pho	1895 (41.4%)	575 (45.5%)	0.33	6e-4
Combgap	1691 (37.0%)	321 (25.4%)	-1.05	2e-24
GAGA-Factor	3318 (72.5%)	746 (59.0%)	-1.27	6e-35
Combgap + GAGA-Factor	1548 (33.8%)	281 (22.2%)	-1.12	1e-25
Zelda	2348 (51.3%)	828 (65.5%)	1.16	8e-33
Pho + Zelda	1041 (22.7%)	455 (36.0%)	1.39	1e-37
GAGA-Factor + Zelda	2002 (43.8%)	617 (48.8%)	0.41	2e-5

Table 1

Enrichment of Selected Transcription Factors with E(z) peaks that accumulate H3K27me3 by late NC14.

Peaks of transcription factors (column 1) overlapping with E(z) peaks were counted and scored for whether overlapping E(z) peaks acquire H3K27me3 by late NC14. 2x2 contingency tables (factor-by-K27me3 domain membership, for the set of all E(z) peaks) were calculated and subjected to Fisher's Exact Test. The number of factor peaks overlapping any E(z) peak (n = 4576) is shown in column 2. The number of factor peaks overlapping E(z) peaks within PcG domains (n = 1264) is shown in column 3. Log2(Odds Ratios) are reported in column 4, and p-values are reported in column 5. Negative log2(odds ratios) indicate depletion (red) and positive log2(odds ratios) indicate enrichment (blue) of a factor associating with E(z) peaks that accumulate H3K27me3. Additional pairwise comparisons that did not yield significant p-values are not shown.

precludes the maturation of H3K27 residues to a fully trimethylated state until mid- to late-NC14, and that PcG factors access chromatin during cleavage divisions to catalyze sub-trimethylated states. To further evaluate this model, we tested for evidence of PRC1 interactions with PcG domains, performing ChIP-seq at NC13 and NC14 (mid- and late-) against H2Aub, catalyzed by the Sce/dRing component of PRC1. In contrast to the H3K27 modification states, H2Aub is readily detected near E(z) peaks and within PcG domains as early as NC13 (**Figure 1A and C**). These results suggest that both PRC2 and PRC1 access and modify *Drosophila* chromatin prior to large-scale ZGA despite rapid cell cycle progression.

Examination of the patterns with which H3K27me3 comes to be deposited at individual domains over the course of NC14 suggests that emergence of this state is produced by a “nucleation followed by spreading” mechanism that initiates at E(z) peaks within PcG domains, which we will refer to as prospective PREs (pPRE). **Figure 1D** shows an example of a representative moderate-sized PcG domain containing the early zygotic gene *intermediate neuroblasts defective (ind)*. The major pPRE in this domain overlaps the *ind* transcription start site, and overlaps with peaks for E(z), Pho, and Cg. The earliest accumulation of H3K27me3 observed at mid-NC14 is always flanking the pPRE and thereafter is seen to ‘spread’ distally.

H2Aub in general at equivalent timepoints demonstrates a greater degree of spreading over the PcG domain distal to sites of presumptive nucleation compared with H3K27me3, suggesting that the ubiquitylation rate of Sce outpaces the trimethylation rate of E(z). Similar kinetics are observed over the PcG domain corresponding to the bithorax complex (**Figure 1E**). Compared with *ind*, the bithorax complex contains numerous sites of nucleation, however, despite the increased density of nucleation sites, H3K27me3 accumulation at the bithorax complex shows a similar delay until mid-NC14.

PRE-binding factors are excluded from nuclei during early cleavage stages

If pPREs were to serve as sites for maintenance of maternal PcG modification states, then effector proteins should localize to nuclei during the cleavage divisions. Protein expression kinetics for PcG components including E(z) and Pho have previously been measured using transgenically expressed fluorescent protein chimeras driven by heterologous alpha-tubulin regulatory sequences (39). However, this study did not address whether nuclear concentration of these factors is consistent over the cleavage divisions. Whereas some maternally supplied proteins such as Zelda readily localize to the nucleus throughout the cleavage divisions (33,40), other key factors, such as Polycomb, Dorsal, Bicoid, TATA-binding protein, Heat Shock Factor, and the large subunit of RNA Polymerase I are subject to regulated nuclear import and/or translation during these stages (29,41–44), demonstrating changes in the fraction of nuclear localized protein as embryos approach ZGA.

We wished to reproduce and extend these prior measurements using endogenously tagged PcG components and therefore we performed live imaging of our GFP chimeras to quantify relative GFP intensity for each chimera spanning from interphase of NC10, when nuclei first migrate to the embryonic cortex, through NC14, when embryos begin gastrulation (**Figure 3**). EGFP-E(z) localizes to the nucleus during interphases 10–14 and, consistent with previous reports (39), is evicted from chromatin during mitosis (**Figure 3A**, Supplemental Movie 1). However, E(z) concentration decreases in each subsequent interphase, reaching a minimum observed nuclear concentration at NC14 when H3K27me3 accumulates on chromatin. Such high initial concentrations could potentially facilitate rapid maturation of H3K27 through consecutive methyl-states despite an expected slow catalytic rate for the me2 to me3 modification (36,37).

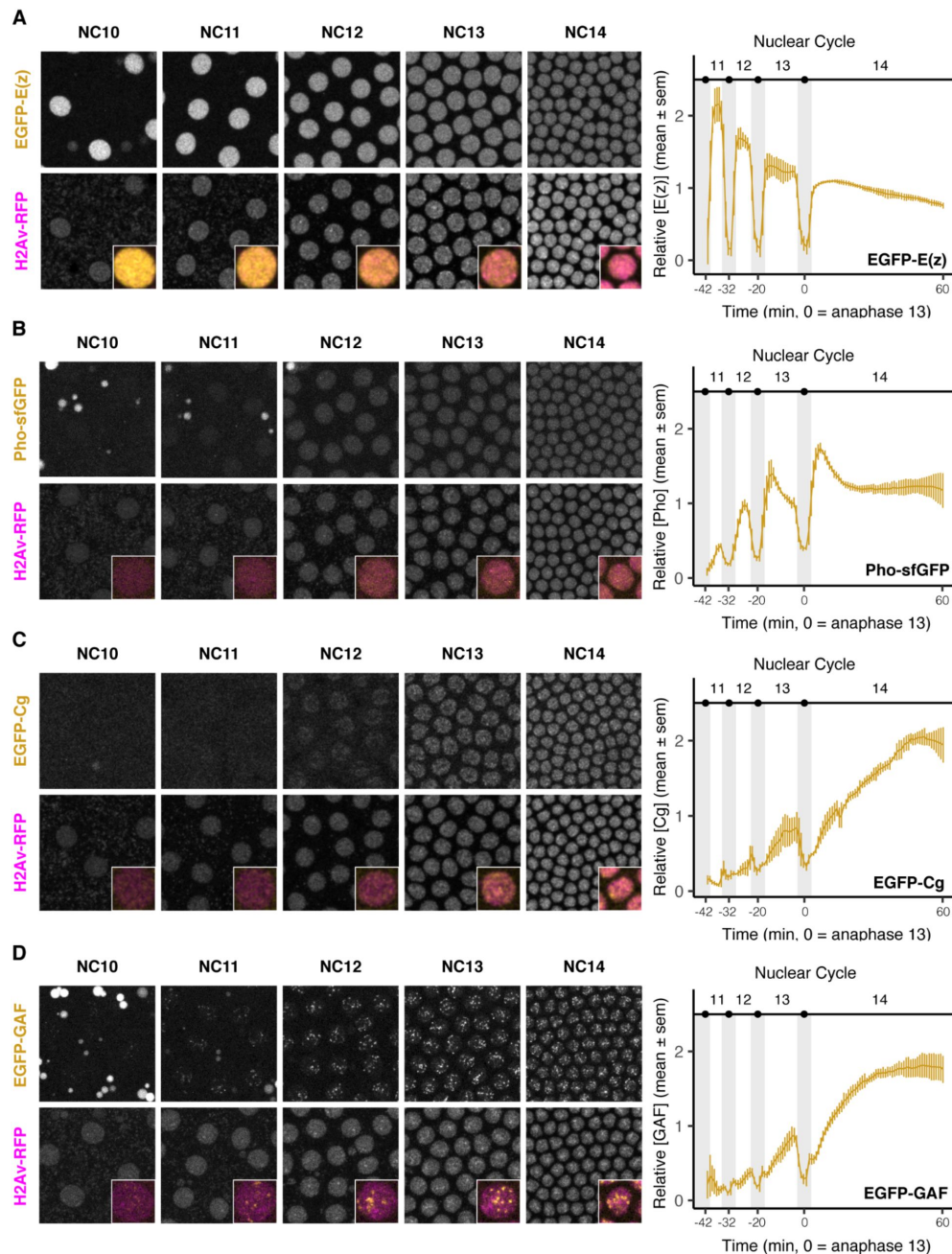


Figure 3

Nuclear localization patterns of E(z) and three PRE-binding factors over late cleavage divisions.

EGFP-E(z) (A), Pho-sfGFP (B), EGFP-Cg (C), and EGFP-GAF (D) were subjected to time-lapse confocal microscopy from nuclear migration (NC10) through the period of cellularization (NC14 + 60 minutes). All specimens also expressed Histone H2Av-RFP to visualize nuclei. The images at left show maximum-projected still images from representative time-lapse movies for the indicated GFP chimera (top) and His2Av-RFP (bottom) at interphase of the indicated nuclear cycles. The insets show a color merge of one nucleus at each respective nuclear cycle (yellow: GFP chimera, magenta: His2Av-RFP). The quantifications at right show the relative nuclear concentration of each GFP chimera from NC11 through NC14 (top x-axis). Grey vertical bars indicate the approximate mitotic periods for each nuclear cycle. Data represent averaged intensity measurements from $n \geq 3$ embryos per chimera, mean normalized $\pm$ standard error of the mean. While E(z) readily localizes to nuclei throughout the cleavage divisions, the three PRE-binding factors Pho, Cg, and GAF increase nuclear concentration over the last nuclear divisions before cell cycle lengthening at NC14.

In contrast, the three measured PRE-binding factors Pho-sfGFP, EGFP-Cg, and EGFP-GAF gradually increase nuclear concentration between NC11 and NC14 (**Figure 3 B-D**). Pho-sfGFP is detected in nuclei during interphase, beginning at a low but detectable level at NC11 and increasing markedly at NC13 and 14 (**Figure 3B**). Consistent with a prior report, Pho-sfGFP is not strongly detected on mitotic chromatin (Supplemental Movie 2) (39). EGFP-Cg demonstrates a more gradual increase in nuclear concentration than Pho-sfGFP over this period, starting with lower apparent concentration at NC11, and continually increasing in nuclear concentration throughout NC14 (**Figure 3C**). Unlike Pho and E(z), Cg is detected on mitotic chromatin, albeit at a lower intensity than during interphase (Supplemental Movie 3). Finally, consistent with previous reports, EGFP-GAF localizes from NC10-14 to bright apical puncta corresponding to simple (AAGAG)_n satellite repeats (25, 45, 46), and nucleoplasmic concentrations of GAF gradually increase over this period, reaching a maximal intensity at NC14 (**Figure 3D**, Supplemental Movie 4). These observations indicate that these three PRE-binding factors are effectively absent from early cleavage-stage chromatin and significantly increase their nuclear concentration over the timeframe of ZGA.

One limitation of live-imaging of GFP-tagged chimeras are potential discrepancies between protein expression and GFP maturation rates, which could lead to underestimates of protein expression levels. To validate the live-imaging measurements, we performed immunostaining on formaldehyde-fixed preparations of our endogenously GFP-tagged E(z), Pho, Cg, and GAF chimeric embryos (**Figure 4**). In accordance with the live imaging results, E(z) was found to localize to the nucleus at all observed nuclear cycle interphases, maintaining an apparent high concentration prior to nuclear migration and blastoderm formation at NC10 (**Figure 4A**). Also in accordance with the live imaging measurements is the observation that neither Pho (**Figure 4B**), Cg (**Figure 4C**), nor GAF (**Figure 4D**) achieve above-background nuclear concentrations until NC11 (Pho) or NC12 (Cg, GAF). Taken together, our live and fixed embryo analysis of E(z), Pho, Cg, and GAF nuclear localization indicates that while E(z) readily localizes to nuclei throughout the cleavage period, at least three PRE binding factors, Pho, Cg, and GAF show only limited nuclear localization until just prior to large-scale ZGA. Although supra-NC14 concentrations of E(z) are observed within nuclei during all syncytial interphases, key factors for mediating nucleation of PcG domains during maintenance phases are apparently absent from chromatin until just before H3K27me3 becomes detectable by ChIP at NC14.

H3K27me3 and H2Aub are not maintained on chromatin at high levels during cleavage divisions

To determine to what degree PcG-dependent chromatin modifications are maintained on cleavage stage chromatin, we next performed immunostaining for PcG-dependent histone modifications H3K27me3 and H2Aub (**Figure 5**). By this approach, we confirm and reproduce the previously reported observation of heterogeneous staining of H3K27me3 in zygotic nuclei shortly after fertilization (**Figure 5A**, NC2) (31). Both nuclei in **Figure 5A** show intense anti-H3K27me3 immunoreactivity over approximately half of each nucleus, and prior observations trace the source of this heterogeneity to the maternal pronucleus, which is marked with significant amounts of H3K27me3, in contrast to the paternal pronucleus, which carries no H3K27me3 marks (31). However, in contrast to prior reports, the H3K27me3 signal becomes undetectable in later nuclear divisions (**Figure 5A**, NC8, NC13) recovering intensity by mid NC14 (**Figure 5A**, NC14). We also measured H2Aub in fixed embryos. Unlike H3K27me3, H2Aub is not strongly detected in nuclei immediately after fertilization (**Figure 5B**, NC3). We do detect, however, a low-level H2Aub signal on the condensed meiotic polar body that is barely above the cytoplasmic background signal (**Figure 5B**, NC10 asterisk) indicating that a small quantity of H2Aub is deposited on maternal chromatin. Even on compacted mitotic chromatin, H2Aub signal is undetectable prior to syncytial blastoderm stages (**Figure 5B**, NC3 & NC8). Coincident with the observed nuclear localization of PRE-binding factors just prior to ZGA, we begin to observe above-background

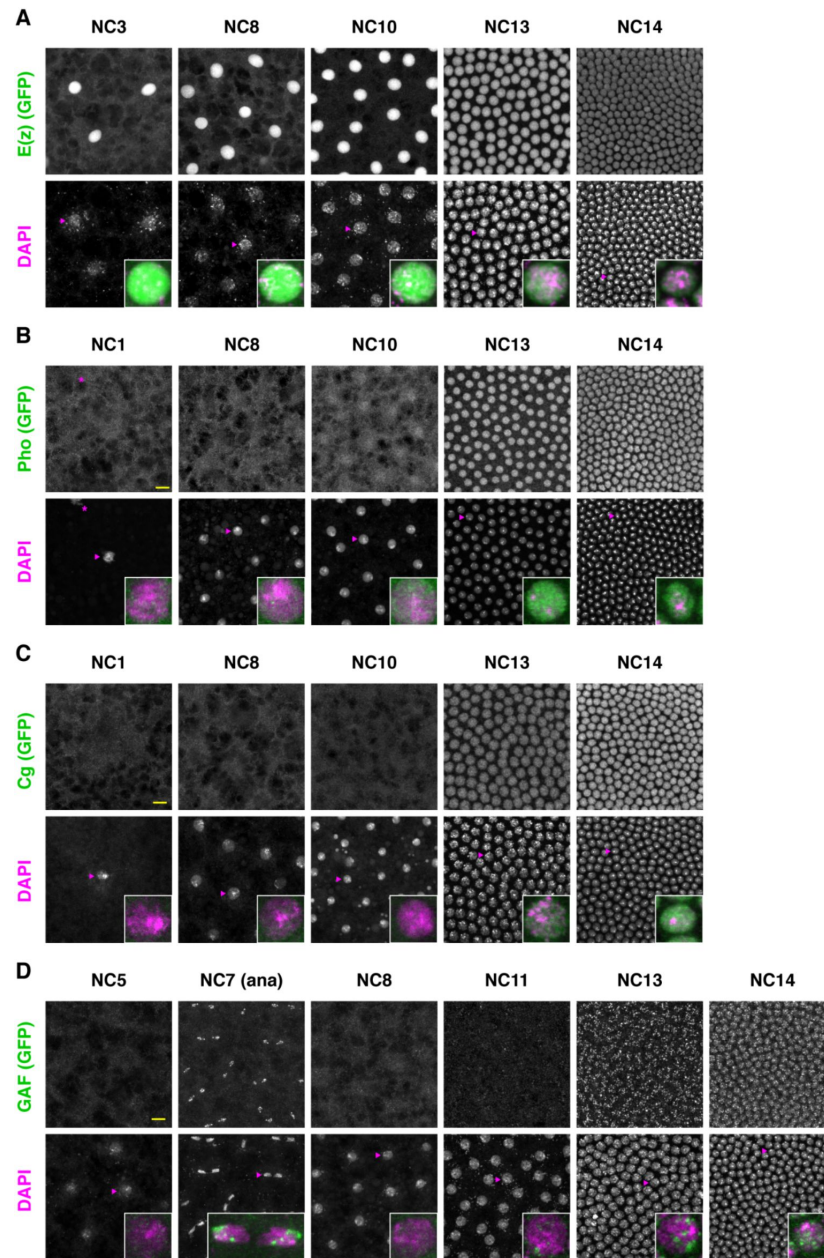


Figure 4

Immunostaining of E(z) and three PRE-binding factors.

Embryos expressing EGFP-E(z) (A), Pho- sfGFP (B), EGFP-Cg (C), and EGFP-GAF (D) chimeras were formaldehyde-fixed and subjected to immunostaining with an anti-GFP antibody to facilitate measurement of nuclear concentration prior to nuclear migration. All images are maximum z-projections. For each chimera, the top row of images show GFP immunofluorescence signal from representative specimens staged at the nuclear cycle indicated at top. Samples were co-stained with DAPI to visualize DNA/nuclei. The insets show an enlarged single nucleus indicated by the magenta arrowhead (green: GFP chimera, magenta: DAPI). Displayed image intensities are consistent for post-migration nuclear cycles (NC10-14) and are linearly adjusted based on DAPI intensity for pre-migration nuclear cycles, when nuclei are found deep within the embryo. For EGFP-GAF (D), one example of a pre-migration mitosis is shown (NC7 anaphase) to demonstrate the ready detection of GAF on mitotic chromatin in contrast to much lower signal during pre-migration interphases. E(z) localizes to interphase nuclei from the beginning of the cleavage divisions and is readily detected. Pho, Cg, and GAF do not localize to interphase nuclei until after nuclear migration. The scale bar is shown in yellow in the bottom right of the top-left image in each panel, size = 10 μm .

H2Aub accumulation by NC12, and this signal increases throughout NC14 (**Figure 5B** [↗](#), NC12 & NC14). These results indicate that whatever degree of H2Aub is inherited maternally, it is not strongly maintained during the cleavage divisions. We conclude that the H2Aub modification is established wholly *de novo* shortly before ZGA. Taken together, these observations are consistent with a model where the initial, maternally determined H3K27 state is diluted by the rapid syncytial divisions, and re-establishment of this modification state is restricted to late cleavage divisions by limiting nuclear localization of nucleating factors such as Pho, Cg and GAF.

Loss of GAGA-factor has limited impact on H3K27me3 establishment at ZGA

Based on the above observations, we next wanted to evaluate the impact of interfering with nucleating factors on the establishment of H3K27me3 patterns at ZGA. We attempted to perform maternal knockdowns of EGFP-Cg and Pho-sfGFP but consistent with prior reports, loss of function for either factor during oogenesis fails to yield significant quantities of viable embryos that develop to cellular blastoderm stages, thus limiting our ability to test this hypothesis with either factor ([47](#) [↗](#), [48](#) [↗](#)). Alternatively, we tested whether loss of GAGA-factor (GAF) would impact the re-establishment of H3K27me3 at ZGA. GAF, encoded by the *Trithorax-related* gene, has previously been implicated as a binding factor at PREs ([49](#) [↗](#)–[52](#) [↗](#)). GAF is amongst a cohort of transcription factors including CLAMP and Pipsqueak that bind GA-rich ((GA)*n*) motifs that are widespread in the *Drosophila* genome and are frequently found within PRE sequences ([53](#) [↗](#), [54](#) [↗](#)). Mutation of (GA)*n* motifs within PRE reporter constructs demonstrates that such motifs can be necessary for PRE-dependent reporter silencing ([12](#) [↗](#), [14](#) [↗](#), [49](#) [↗](#)–[52](#) [↗](#)), and by ChIP-seq GAF is observed to bind 60% of pPREs ([55](#) [↗](#)). Given prior reports of GAF's role for pioneering chromatin accessibility at ZGA and its expected role for PRE-dependent gene silencing ([25](#) [↗](#)), we wished to test whether GAF function is required for the initial establishment at ZGA of H3K27me3 within PcG domains.

To knock down maternal GAF in early embryos, we turned to the recently developed Jabba-trap system (JT, ([56](#) [↗](#))) to sequester an N-terminal EGFP-tagged GAF to cytoplasmic lipid vesicles (see Materials and Methods). Expression of JT using maternal alpha-tubulin 67c regulatory elements sequesters EGFP-GAF away from nuclei in cleavage stage embryos both during mitosis and interphase (**Figure 6A** [↗](#), Supplemental Movies 5 and 6). By ChIP-seq, JT-mediated knockdown of EGFP-GAF (JT-GAF) results in a nearly complete depletion of GAF from blastoderm stage chromatin (**Figure 6B** [↗](#)), correlating well with the degree of cytoplasmic sequestration observed by microscopy. Given the extensive depletion in JT-GAF embryos, we next measured H3K27me3 to evaluate the dependency of this modification on GAF function.

Despite widespread binding of GAF to pPRE sequences and a prior expectation of GAF's necessity for PRE function, we observe extensive establishment of H3K27me3 following JT-mediated knockdown of GAF and little major change in the modification status of pPREs (**Figure 6C-G** [↗](#)). A general challenge for quantitative evaluation of the impact of factor knockdown experiments on the establishment of H3K27me3 is the discrepancy in length scales at which transcription factors and PcG modifications interact with genomic loci. To quantify the effect of JT-GAF on H3K27me3, we counted recovered H3K27me3 ChIP-seq reads in 2 kb bins spanning the genome for replicates (*n* = 3) of each genotype and tested for differential enrichment between wild type (EGFP-GAF) and knockdown (JT-GAF) conditions using DESeq2 ([57](#) [↗](#)) (see Materials and Methods and Supplemental Material). This allowed us to score whether any of the 2 kb bins within PcG domains (*n* = 3883) showed significant changes in H3K27me3 distribution following GAF knockdown. By this approach, we observe only a limited number of 2 kb bins within PcG domains with modest decreases in H3K27me3 signal following GAF knockdown, and none of these pass significance testing (adjusted *p*-value < 0.05) at a standard magnitude of effect cutoff (log₂(Fold Change) < -1) (Supplemental Figure 3). A greater number of 2 kb bins within PcG domains show increases in H3K27me3 signal following GAF knockdown, and eight (0.2%) of these bins are scored as

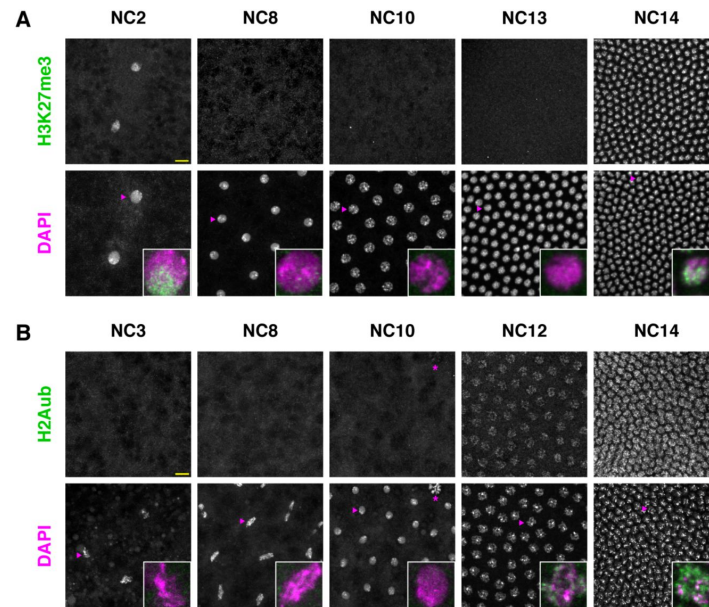


Figure 5

Immunostaining of H3K27me3 and H2Aub during cleavage stages.

Embryos (w^{1118}) were fixed and subjected to immunostaining for H3K27me3 (A) and H2Aub (B). Samples were co-stained with DAPI to highlight DNA/nuclei. Each panel shows maximum z-projected immunostaining for the indicated histone modification in the top row of images at the indicated nuclear cycle (top). The bottom row shows the corresponding DAPI image. Image intensities are normalized as described for [Figure 4](#) to allow for semi-quantitative comparison of signal intensities across nuclear cycles. The inset shows a single enlarged nucleus (indicated by a magenta arrowhead) with a color merge of the modification (green) and DAPI image (magenta). The magenta asterisk in panel B, NC10 indicates the position of the polar body. H2Aub is not detectable in somatic nuclei above background until late cleavage divisions. The scale bar is shown in yellow and corresponds to 10 μ m in all cases.

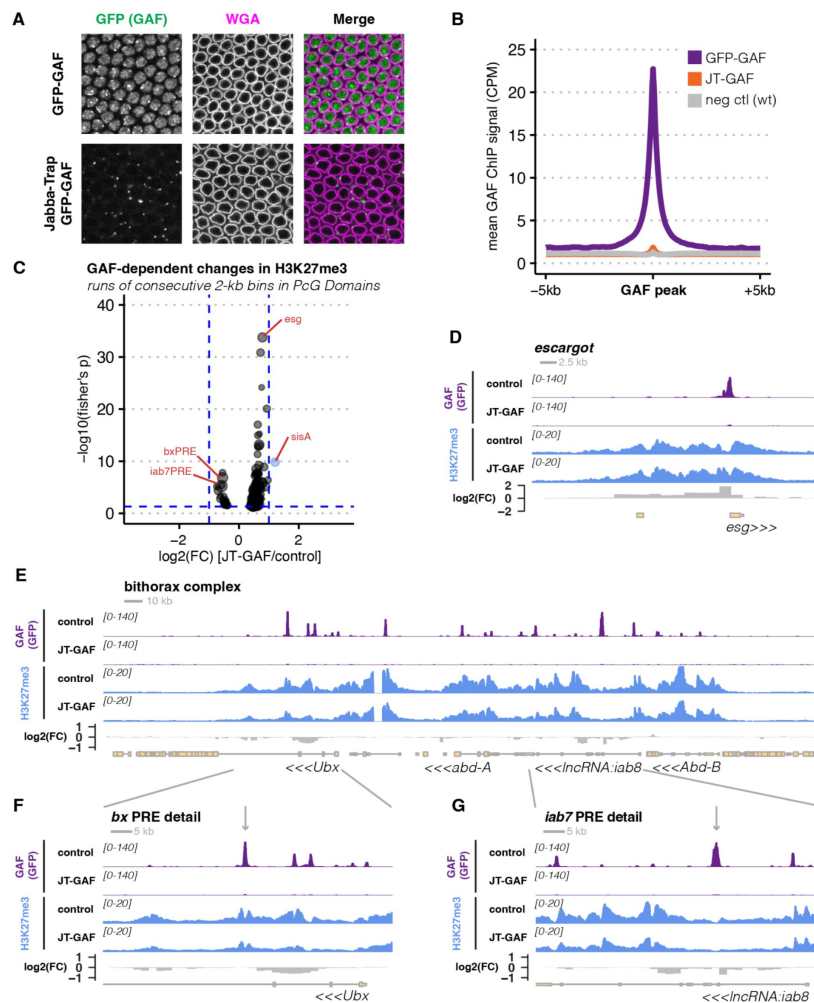


Figure 6

GAF knockdown has limited impact on accumulation of H3K27me3 at NC14.

A) Control (EGFP-GAF) or GAF-knockdown (Jabba-Trap/+; EGFP-GAF) embryos were formaldehyde fixed and immunostained for GFP to detect GAF at NC14. Specimens were counterstained with wheat germ agglutinin (WGA) to highlight nuclear membranes. A single z-slice of individual channels (grey) and merged channels (GAF: green, WGA: magenta) shows that in the presence of Jabba-trap, EGFP-GAF is undetectable within NC14 nuclei. B) ChIP-seq was performed against EGFP-GAF in either positive control (EGFP-GAF), Jabba-trap GAF (JT-GAF), or negative control (*w¹¹¹⁸*) samples with an anti-GFP antibody. Following peak calling, average GAF ChIP-seq signal in counts per million (CPM) over GAF peaks was measured and plotted, with ± 5 kb flanking sequence. The plot shows that whereas GAF (purple) is strongly enriched over peaks, Jabba-trap (orange) reduces GAF occupancy to near-background (grey) levels. C) A volcano plot is shown for the results of a DESeq2 differential enrichment analysis for H3K27me3 ChIP-seq between control (EGFP-GAF) and JT-GAF. Enrichment was measured over 2 kb bins and ‘runs’ of consecutive bins with p-values < 0.05 were merged. The volcano plot shows on the x-axis the average $\log_2$ fold change (FC) of the merged bins and the $-\log_{10}$ value of adjusted p-values merged using Fisher’s method. The size of a plotted point indicates the relative mean counts for the run of bins. Blue dotted lines indicate traditional cutoffs for significance testing ($\text{abs}(\log_2(\text{FC})) > 1$ & $p\text{-adj} < 0.05$). Runs of bins exceeding these thresholds are colored blue. Regions of interest are labeled in red. While few regions show large-magnitude, GAF-dependent changes in K27me3, the general trend is for this modification to increase within PcG domains in the absence of GAF. D) Detail of ChIP-seq coverage over the *escargot* PcG domain. GAF ChIP seq for control or JT-GAF is shown at top (purple) and H3K27me3 for both genotypes is shown at bottom. The $\log_2(\text{FC})$ from the DESeq analysis over 2 kb bins is plotted in the bottom track. Scale bar is shown at top left, and y-axis range for coverage plots are indicated in brackets, with units of counts per million. E-G) Detail of ChIP-seq coverage over the bithorax complex (E) and zoomed-in views of the *bx* (F) and *iab7* (G) PREs are shown. Tracks are as described for panel D. The location of the PREs in F and G are indicated with a grey arrow. Scale bars are shown at top left for each respective plot. Y-axis range for coverage plots are indicated in brackets with units of counts per million.

significant increases by the aforementioned criteria (Supplemental Figure 3). Closer visual examination of H3K27me3 coverage identifies regions within PcG domains that may have functionally relevant, albeit partial, changes in H3K27me3 following GAF knockdown, where the magnitude of effect may not reach typical thresholds for significance calling. To identify such regions, we de-emphasized log₂(Fold Change) as a criterion for significance calling and identified consecutive genomic runs of 2 kb bins within PcG domains with statistically significant differences in H3K27me3 signal (i.e., adjusted p-value < 0.05 with any log₂(Fold Change)), yielding 361 runs of bins ranging in size from 2 to 24 kb in size (Supplemental Figure 3, Materials and Methods). By this approach, 48% of PcG domains (n = 124) contain one or more runs of 2 kb bins with modest localized changes in H3K27me3 following GAF knockdown. Two of these domains have mixed changes, containing runs of bins that both show modest increases and decreases in H3K27me3 signal, the remainder show either increases or decreases. Of these, only 17 PcG domains (6.7%) are associated with one or more runs (n = 23) that undergo net reductions of this modification.

The bithorax cluster is the major representative PcG domain to undergo reduction in H3K27me3 following GAF knockdown, accounting for four of the 23 runs with net reductions averaging a log₂(Fold Change) of -0.59 within significant 2 kb bins. At the bithorax cluster, we observe GAF to be necessary to attain maximal modification of the regions flanking the *bithorax* and *iab-7* PREs (Figure 6E-G, (12, 52, 58)). These effects of GAF on K27me3 deposition are likely to be direct, as 82% (n = 19) of these runs overlap with one or more GAF ChIP-seq peaks, including all four runs within the bithorax cluster. Effects like this are rare, however, and the major trend in these data is consistent with a role for GAF to counteract deposition of H3K27me3, since loss of GAF results in net gains of H3K27me3 at subregions within 109 PcG domains compared with only 17 domains with reductions. These GAF-dependent increases in H3K27me3 signal are also likely to be direct, as 109 (100%) of the runs contain one or more GAF ChIP peaks. As an example of this effect, the strongest gain in K27me3 signal following GAF knockdown is observed at the *escargot* locus, where an individual 2 kb bin corresponding to a GAF peak and the *escargot* transcriptional start site exceeds a log₂(Fold Change) > 1 increase in K27me3, and runs of 2 kb bins within its associated PcG domain see an average log₂(Fold Change) increase of 0.78 (Figure 6D and Supplemental Figure 3). Taken together, these observations suggest that loss of GAF in blastoderm stage embryos results in limited effects on the establishment of H3K27me3, and that most of these moderate effects are consistent with a model where GAF operates at ZGA to counteract acquisition of high levels of this modification.

Zelda is necessary for licensing a subset of PcG domains to deposit H3K27me3

The zinc-finger transcription factor Zelda is a major activator of the zygotic genome, operating through several known regulatory mechanisms (24). As a pioneer factor, Zelda is necessary for establishing and maintaining chromatin accessibility at a cohort of genomic loci required for the expression of a subset of early zygotic genes (59–62). As a transcription factor, Zelda also regulates gene expression, functioning mainly to activate target gene transcription (33, 63–65). Studies on *zelda* mutant embryos have also demonstrated additional phenotypes, including effects on genomic compartmentalization and formation of transcription factor hubs, that could either stem from pioneering or trans-regulatory activities or could represent additional biochemical functions (66–69). Given Zelda's role for activating expression of a cohort of zygotic genes, including those associated with PcG domains, we wished to measure the effect of Zelda on the establishment of PcG domains. We reasoned that Zelda could impact the modification state of these domains in at least two opposing ways. As an activator of transcription, Zelda could be required to promote an active state and suppress establishment of PcG modifications at target genes. Alternatively, as a pioneer, Zelda could be required to delineate cis-regulatory elements required for the establishment of PcG domains and modification of target genes. In the first hypothesis, loss of *zelda* would be predicted to result in the gain of PcG chromatin modifications at the expense of modifications associated with transcriptional activity. In contrast, with the second

hypothesis, we would expect to observe in *zelda* mutants a loss of PcG modifications, independent of the transcriptional status of a locus. We therefore sought to distinguish these competing mechanisms.

To distinguish these hypotheses, we performed ChIP-seq for a battery of targets associated with transcriptional activity (RNA Pol2 (CTD pSer5)) and repression (E(z), H3K27me3, and H2Aub) comparing wild type embryos with those produced by *zlda*²⁹⁴ germline clones staged precisely at mid-NC14 (see Materials and Methods). Over the set of PcG domains in *zelda* mutants, we observe both increases and decreases in PcG histone modifications. As above, we quantified *zelda*-dependent changes in H3K27me3 ChIP-seq signal over 2 kb genomic bins. Unlike for GAF, loss of Zelda yields several genomic tracts with large-magnitude changes in H3K27me3. Of the 3746 bins located within PcG domains for this analysis, 6% (n = 226) show statistically significant reductions in H3K27me3 at a log2(FoldChange) threshold of -1 (Supplemental Figure 4). A smaller number (n = 62, 1.7% of bins in PcG domains) show increases in H3K27me3 signal following loss of *zelda* (Supplemental Figure 4). Taking consecutive runs of bins with statistically significant adjusted p-values (independent of log2(Fold Change)), we recover 546 runs within PcG domains ranging in size from 2 to 46 kb in size. Of these, 55.1% (n = 301) show moderate to large increases in H3K27me3 signal, with 1.6% (n = 9) exceeding an average log2(FoldChange) threshold of 1 indicative of a large magnitude of effect. The remaining 44.9% (n = 245) of runs show decreases in H3K27me3 signal, with 5.3% (n = 29) exceeding an average log2(Fold Change) threshold of -1 indicating a large magnitude of effect (**Figure 7A**). We observe a high degree of correlation between Zelda-dependent changes in H3K27me3 and H2Aub (spearman's rho = 0.78, Supplemental Figure 4). The role of Zelda within PcG domains is therefore complex, with Zelda activity counteracting establishment of PcG modifications at a small number of domains, and Zelda activity being required for establishment at others.

We next compared how changes in H3K27me3 status correlated with changes in RNA Pol2 recruitment following loss of Zelda. Depending on biological context, when early zygotic loci fail to be activated in *zelda* mutant embryos, a subset of these sites undergo changes in H3K27me3 status. As previously reported, we observe a significant loss of RNA Pol2 occupancy in *zld* mutant embryos at a subset of early zygotic target genes (70), and similar losses in histone modifications associated with active transcriptional states are also observed (28). Through DESeq2 analysis, of 798 early, zygotic-only transcribed mRNA-encoding loci (30), 269 (33%) show statistically significant reductions of RNA Pol2 occupancy in *zelda* mutant embryos. Within this set of zygotic-only genes, 155 (19%) reside within a PcG domain, and 43% (n = 66) of these require Zelda for full RNA Pol2 occupancy. Comparing correlations between Zelda-dependent losses in RNA Pol2 occupancy and changes in H3K27me3 deposition, while there are 66 early zygotic-only genes within PcG domains that require Zelda for RNA Pol2 occupancy, only 14 (21%) of these show significant changes in H3K27me3, and of these, two account for increases, and 12 account for decreases in H3K27me3 deposition (**Figure 7B**). Therefore, while Zelda affects H3K27me3 establishment at a subset of targets within PcG domains, not all sites requiring Zelda for RNA Pol2 recruitment show changes in this modification. Of the subset of targets with Zelda-dependent changes in both H3K27me3 and RNA Pol2, most sites demonstrate losses in occupancy of both factors following loss of Zelda.

We also tested for evidence that Zelda could be required to counteract establishment of PcG modifications at genes not associated in wild type embryos with a PcG domain. As described above, E(z) binds near gene promoters located both within and outside of PcG domains. All 66 of the genes within PcG Domains that require Zelda for full RNA Pol2 binding also coincide with a promoter-proximal E(z) peak. The majority (134/ 203, 66%) of zygotic-only genes outside PcG domains that require Zelda for RNA Pol2 occupancy also coincide with a promoter-proximal E(z) peak. Despite this, loss of Zelda does not result in significant acquisition of H3K27me3 at genes located outside of PcG domains. Of the 203 genes requiring Zelda for RNA Pol2 recruitment, only one, *Ocho*, shows a significant increase of H3K27me3 over its transcription start site upon loss of

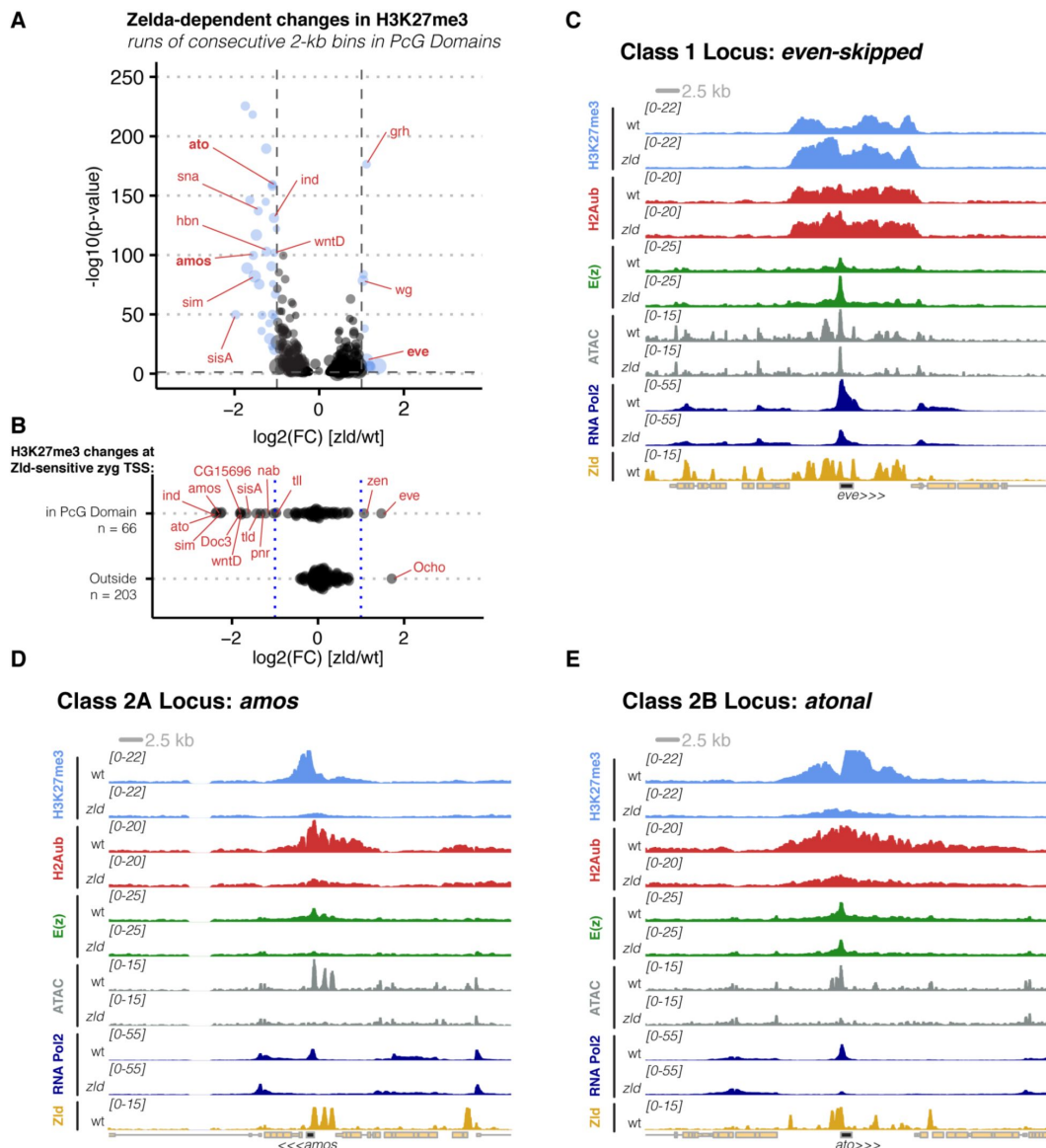


Figure 7

Zelda is necessary during ZGA for the establishment of H3K27me3 at a subset of PcG domains.

A) A volcano plot is shown for DESeq2 analysis of H3K27me3 ChIP-seq comparing wild type and *zelda* germline clone embryos. Plotting methodology is as described in [Figure 6C](#). On average, Zelda activity is required for establishment rather than antagonism of H3K27me3 at a subset of PcG domains. B) H3K27me3 is predominantly lost at sites requiring Zelda for RNA Pol2 recruitment. The $\log_2(\text{FC})$ of H3K27me3 ChIP-seq is plotted for 2 kb bins containing TSS that depend on Zelda for Pol2 recruitment. In the absence of Zelda, the TSS included in this analysis have major reductions in Pol2 occupancy. TSS were grouped into two categories: those within a PcG domain (top) and those outside a PcG domain (bottom). In the absence of Zelda, TSS within PcG domains that lose Pol2 ($n = 66$) also typically see losses in H3K27me3, with the exception of a few loci (*eve*, *zen*) that see increases. Loss of Pol2 at Zelda-sensitive TSS outside of PcG domains ($n = 203$) does not correspond to *de novo* accumulation of H3K27me3, except at the *ocho* locus. C-E) Coverage plots for loci representing classes of PcG domain responses to *zelda* loss of function. ChIP for factors indicated at left were performed for wild type (top track) and *zelda* germline clone (bottom track) embryos. ATAC data are from (Hannon 2017) and Zelda ChIP-seq is from (Harrison 2011). Scale bar is at top left and the y- axis range for each displayed track is indicated in brackets, with units of counts per million. Panel C shows a class 1 locus, *even skipped*. Class 1 are PcG domains where increases in K27me3 are seen in the absence of *zelda*. Panel D and E show Class 2 loci, which are PcG domains where decreases in K27me3 are seen in the absence of *zelda*. Class 2A loci (panel D) see reductions in K27me3 accompanied by loss of E(z) recruitment. Class 2B loci (Panel E) see reductions in K27me3 without significant loss of E(z) recruitment.

Zelda (**Figure 7B** [↗](#)). Therefore, these Zelda-dependent effects on H3K27me3 deposition are almost entirely limited to the territories of PcG domains, and Zelda does not counteract in early embryos a latent competency for additional genomic loci to acquire PcG modifications. These results indicate that Zelda influences the establishment of chromatin modifications within a subset of PcG domains and allows us to categorize *zelda*-sensitive domains into two major classes. Class 1 domains (n = 9) demonstrate increases and Class 2 domains (n = 29) demonstrate losses in both K27me3 and H2Aub following loss of Zelda. We focus on the characterization of these two classes of targets in the following analysis.

The increases in PcG modifications within Class 1 domains support a mechanism where Zelda's trans-activation function is necessary to counteract the establishment of H3K27me3 and H2Aub. This class includes PcG domains associated with patterning genes such as *even-skipped*, *wingless* and *grainyhead*. We focus here on *even-skipped* as an exemplar of this class (**Figure 7C** [↗](#)). In *zelda* mutants, the *eve* transcription unit has significantly reduced RNA Pol2 occupancy, and several cis-regulatory elements within the *eve* PcG domain lose chromatin accessibility, but accessibility is retained in a promoter-proximal region that harbors the major peak of E(z) binding within this domain. E(z) occupancy is increased at the *eve* promoter pPRE in *zelda* mutants, and corresponding increases in H3K27me3 and H2Aub are observed within this locus. Notably, these increases are limited to the bounds of the wild type *eve* PcG domain, indicating that boundary integrity at this locus remains intact in the absence of *zelda*. These observations at the *eve* locus, and for Class 1 domains in general, are consistent with a model where loss of a trans-activating factor can result in gains in PcG activity at least at sites that are competent to do so.

In contrast to the Class 1 domains, Zelda is necessary for establishment of PcG modifications within 28 Class 2 domains, where loss of Zelda results in a failure to establish H3K27me3 and H2Aub during NC14. Comparison of chromatin modification states between wild type and *zelda* mutant embryos within the set of Class 2 domains identifies two major sub-classes which we will refer to as 2A and 2B, both of which show extensive loss of PcG modifications, but stemming from different effects of Zelda on chromatin accessibility and E(z) recruitment. In *zelda* mutants, Class 2A domains fail to establish PcG modifications, and this effect correlates with Zelda-dependent losses in chromatin accessibility and E(z) binding. Overall, we identify 6 total Class 2A domains, which contain *sisterless A*, *amos*, *cadherin-N*, *wntD*, *Chronophage*, and *Anion exchanger 2*. We focus here on *amos* as the exemplar of this class (**Figure 7D** [↗](#)). In *zelda* mutants, RNA Pol2 occupancy of *amos* is completely eliminated. Similarly, the three major accessible chromatin regions within this domain are *zelda*-dependent and fail to open in *zelda* mutants. E(z) binding within this domain is correspondingly lost in *zelda* mutants, and H3K27me3 and H2Aub are not established within this domain. In contrast, Class 2B loci (n = 17) demonstrate similar failures in establishment of PcG modifications but retain tracts of accessible chromatin and E(z) binding. Class 2B domains include developmental regulators such as *single-minded*, *snail*, *zfh-1*, *atonal*, *teashirt*, and *ind*. We focus here on *atonal* as the exemplar of this class. Similar to the *amos* domain, the *atonal* locus fails to fully establish H3K27me3, and to a lesser extent, H2Aub. While several prominent accessible chromatin regions within the *atonal* domain are Zelda-dependent, the main E(z) binding site within the *atonal* promoter shows a nearly complete loss of accessibility, but no quantitative difference in overall E(z) binding. Yet, despite this retention of E(z), very little H3K27me3 is deposited at this locus in *zelda* mutants. These observations for Class 2 domains are consistent with a mechanism where Zelda activity is necessary for designating a subset of genomic regions as PcG domains, albeit through different mechanisms. For Class 2A loci, Zelda operates as would be expected for a pioneer factor, namely, to establish accessibility at critical cis-regulatory elements, including presumptive PRE elements. Zelda's role at Class 2B loci is less clear: although E(z), and presumably PRC2, is retained at pPRE sequences, in the absence of Zelda, these sites are not fully licensed to modify the surrounding domain.

Discussion

In this study, we sought to understand the mechanism for re-establishment of the H3K27me3 modification during *Drosophila* ZGA. Prior studies have been inconsistent on whether this modification persists throughout the cleavage divisions, or whether it is largely established following large-scale ZGA (28–32). In addition to repeating prior observations of limited H3K27me3 on late cleavage-stage chromatin and rapid accumulation during NC14, we extend these prior observations by showing that the DNA-binding factors thought to be necessary for nucleation and maintenance of PcG states demonstrate limited nuclear localization until the final few cleavage divisions. These observations are consistent with a model where the rapid cleavage divisions operate to reprogram the zygotic genome by diluting parental histone post-translational modifications which could otherwise influence gene expression states. Histone modifications associated with both active and inactive states are established upon cell cycle lengthening at NC14 (28,30), and it has been argued that, in the absence of gene activation, sites will default to a repressed, PcG-modified state (28,71). Our results argue that establishment of the active-versus-repressed dichotomy requires prior regulatory input from transcription factors such as Zelda.

Here we observe that, at ZGA, E(z) and the nucleating factors Pho, GAF, and Cg bind broadly across the genome, but H3K27me3 deposition is only observed at a small subset of these sites. Prior genome-wide profiling of PcG components has observed a range of concordances between PcG factor localization and domains of H3K27me3 deposition. Early genome-wide studies in cell lines and broadly staged embryos found high degrees of concordance between PcG factors and H3K27me3 deposition (55,72,73). However, more contemporary studies on tissues and different cell lines have found extensive binding of PcG components including nucleating factors outside of H3K27me3 domains (19,74–77), raising the possibility that PcG domains vary as a function of cell fate determination and that PcG binding outside of domains reflect a form of bivalency or poising for potential PcG regulation. Our results indicate that drastically reducing the fraction of active genes at ZGA does not lead to a corresponding increase in deposition of H3K27me3 outside of PcG domains and argues against a simple model where inactive genes default to a H3K27me3 modified state. Rather, this default state is specified. Zelda plays multiple roles within PcG domains to influence the histone modification status: depending on biological context, Zelda is necessary to counteract H3K27me3, pioneer accessibility of key pPRE sites, or license bound PRC1/2 to deposit their respective histone modifications. Zelda could function to counteract H3K27me3 modification status at certain sites by promoting the acetylation of H3K27 through recruitment of the CBP co-activator (78,79). Zelda's role as a pioneer factor is well appreciated, and PRE function is thought to require establishment of accessible chromatin states (60,61,80). At present, the mechanism whereby Zelda could function to license PRC1/2 to deposit histone modifications is unclear, but these results point to a need to understand further the mechanisms whereby genomic loci are specified to partition into distinct classes of epigenetic regulatory compartments. While enrichment of E(z) at sites that require Zelda for licensing does not appear to depend on establishment of accessible chromatin, perhaps accessibility is required for additional cofactors to bind and signal to PRC1/2 to deposit modifications. Zelda has also been implicated in the formation of higher order chromatin structure (69). While loci like *eve* retain compartmentalization of PcG modified states in the absence of Zelda function, perhaps loci requiring Zelda for licensing fail to attain the correct higher-order chromatin structure necessary for efficient establishment of PcG states (81,82). Notably, Zelda's effect on PcG states is limited to a subset of Zelda targets within PcG domains. Zelda-dependent PcG domains are enriched for loci involved in neurogenesis (e.g., *sna*, *amos*, *ind*, *ato*, *hbn*, *sim*, *N-cad*), where Zelda is also required in later developmental stages (83,84). Within the context of embryonic patterning, Zelda could operate at these sites under the context of reprogramming not only to “put them in to play” but also to prime these sites for later silencing in non-neural cell lineages. Additionally, not

all Zelda-bound pPREs depend on Zelda for licensing and not all pPREs contain Zelda binding sites. We speculate that additional factors (or combinations of factors) will function like Zelda to license pPREs to establish the active-versus-repressed regulatory dichotomy at developmental patterning genes.

We sought to test directly the requirement for GAF in the establishment of PcG modifications at ZGA and observe that loss of GAF function has limited effect on the initial distribution of H3K27me3. GAF has been associated with PRE function due to the frequent occurrence of (GA)*n* sites in PREs, and observations of impaired silencing function of PRE reporters when such sites are mutated (49–52). Additionally, GAF has been shown to be required to establish long-range interchromosomal contacts between homologous copies of endogenous and transgenic PREs (85), whose pairing is necessary for silencing function. Given the strong loss of silencing observed at PRE reporters with mutated (GA)*n* motifs, we speculate that loss of maternal GAF function is compensated for by other (GA)*n* binding factors such as CLAMP and Pipsqueak. Alternatively, it is possible that GAF is required during the maintenance but not the establishment phases of PcG domains. Loss of function studies for other PRE-binding factors in later developmental contexts have shown that loss of Pho can strongly reduce not only H3K27me3 maintenance but also binding of PRC1/2 components (19). To address these questions in the context of establishment, robust methods for depleting maternally supplied proteins in the embryo will be required. Future studies will address this issue.

Maintenance of long-term transcriptional repression in proliferative cells by the PcG system depends on PRE function to nucleate the replacement of modified histones following DNA replication (20,21). Here, we repeat prior measurements demonstrating that the H3K27me3 modification fails to persist on chromatin through the cleavage divisions, and that key nucleating factors demonstrate limited nuclear localization until shortly before ZGA. Prior kinetic modeling of E(z) catalysis on cleavage stage chromatin supports this conclusion, with the caveat that this kinetic model severely under-estimates the observed rate of H3K27me3 accumulation at ZGA (38). *Drosophila* and other organisms (31,85–88) demonstrate the capacity to maintain PcG-dependent epigenetic states across generations, in a phenomenon termed transgenerational epigenetic inheritance (TEI). In the case of *Drosophila*, TEI has been most clearly shown through the stable inheritance over dozens of generations of PRE-bearing transgenes entrained either to active or repressed states (88). In the mouse, the H2Aub modification state is propagated through the maternal germline and serves as the mechanism for re-establishment of patterns of PcG modifications at the morula stage of development through allosteric stimulation of PRC2 activity (86,89). PRC1 activity is limited in the *Drosophila* ovary (90), and consistent with this, we do not observe significant H2Aub on cleavage stage chromatin until shortly before ZGA, thus limiting its potential role in the context of *Drosophila* TEI. If mechanisms for TEI follow the same rules as maintenance of PcG states in late embryos, larvae, and adult tissues, then the observed lack of key nucleating factors until late cleavage divisions should limit the extent to which maternal H3K27me3 states are directly maintained (20,21). Further investigation is necessary to determine the mechanism whereby parental PcG states can be inherited across the maternal to zygotic transition.

Limitations of the Study

We acknowledge the following limitations to this study. Given the nature of antisera and immunodetection, we cannot completely rule out the action of low level (undetectable) but biologically significant quantities of Pho, GAF, Cg, or even H3K27me3 operating at early cleavage divisions. However, whatever level of these factors may be present in early cleavages, their nuclear abundances increase many-fold by NC14. The strength of this quantitative conclusion is driven by our use of GFP chimeras for the measurements of PcG effector proteins, which allows us to use a single antibody source and staining/imaging conditions, thus limiting the expected

variability between antisera raised against distinct targets. We also acknowledge that while GAF knockdown by Jabba-trap eliminates nearly all detectable GAF on chromatin, it remains possible, however unlikely, that the very little GAF that remains associated with PREs is sufficient to promote H3K27me3 establishment at ZGA and can account for the reported limited effect of GAF knockdown on this modification.

Materials and Methods

Drosophila Stocks and Gene Editing

Drosophila stocks were maintained on an enriched high-agar cornmeal media as reported previously (91). Embryos were collected from small population cages on apple-juice agarose plates smeared with a small amount of active yeast paste.

The wild type genotype for this study is *w[1118]* and serves as the ultimate genetic background for gene edited stocks. CRISPR/Cas9 was used to generate fluorescent protein chimeras of *E(z)*, *Pho*, and *Combgap*. The Cas9 transgenic stock used for genome editing for *E(z)* and *pho* was *y w; nos>Cas9/CyO. cg* genome editing was performed in the *y vas>Cas9 ZH2A w* background. Injections were performed in-house as described previously (92).

Combgap

Of the four GFP chimeras used in this study, *cg* has the most complex splice-variant structure and current gene models identify no constant N- or C-terminal coding sequence. The N-terminus, however, only has two predicted splice isoforms, both originating from a common transcription start site, with one minor splice variant (isoform C) omitting the otherwise canonical translation start site. We decided to generate a *cg* allele that would produce *cg* mRNA variants with a common start codon by eliminating alternative start codon usage of isoform C variants. To do this, we mutated the splice acceptor that distinguishes the isoform C splice variant (chr2R:14174461-14174461 (G>A)) and we introduced a *Drosophila* Codon Optimized EGFP sequence plus an 11 amino acid linker via homology directed repair to replace the canonical start codon. Transcripts produced from this *EGFP-cg* locus would either contain the entire *Start-EGFP-linker-cg* sequence, or isoform C splice variants would now splice-in to the EGFP cassette at a downstream cryptic splice acceptor site upstream of EGFP glycine 4, and fail to initiate translation at the intended methionine 1. We therefore subsequently replaced glycine 4 with a methionine residue to allow for isoform C variants to initiate translation at the N-terminus of the EGFP cassette. We sequenced *EGFP-cg* cDNAs via Sanger sequencing to verify that all spliced isoforms contained an N-terminal EGFP cassette with a functional start codon. Genome-edited *EGFP-cg* flies are homozygous-viable.

cg gRNA target: chr2R:14174457-14174476 (-) 5'- GAAGGGCGGCGGATTCTGGG (CGG) -3'

The underlined bases in the *cg* gRNA target indicate sites where synonymous mutations were introduced to eliminate cutting of the homology directed repair template (G>A in the PAM sequence), and where the splice acceptor for isoform C was mutated as described above.

The homology repair template contained 500 bp of left- and right-homology regions upstream of the EGFP-linker cassette and downstream of the mutation to the isoform C splice acceptor site. The homology repair template was constructed in a modified pBluescript vector with a 3xP3>dsRed cassette in the vector backbone that could be used to screen out unwanted whole-plasmid integrants produced by homologous recombination. The gRNA was expressed using the pU6 BbsI vector. Likely transformants were identified by measuring the frequency of mutations induced by *ebony* co-CRISPR as described previously (92).

Individual edited chromosomes were identified using PCR and correct site integration was verified by Sanger sequencing.

E(z): E(z) is expressed with a single common start codon. Prior work (39) determined that E(z) could tolerate an N-terminal EGFP fusion. We therefore targeted the N-terminus with a multifunctional cassette, *start-EGFP-AID-3xHA-linker*, that would facilitate visualization, conditional knockdown, and immunodetection/affinity purification through the EGFP, auxin inducible degron (AID), and 3xHA epitope tag components. Genome-edited *EGFP-E(z)* flies are homozygous-viable. In experiments not presented in the current manuscript, we verified that EGFP-E(z) could be targeted by Jabba-Trap to yield maternal-zygotic E(z) loss of function phenotypes at high frequency and penetrance. Additionally, we determined that the auxin inducible degron tag inserted at this position was not sufficient to yield knockdown of E(z).

E(z) gRNA target: chr3L:10637753-10637772 (+) 5'- GCTATTCATAATGCCTTCGA (GGG) -3'

The underlined bases indicate the position of the *E(z)* start codon on the minus strand. Because the gRNA straddles the insertion site at the start codon, no additional mutations were introduced to prevent targeting of the homology directed repair template.

The homology repair template contained 500 bp of left- and right-homology regions upstream and downstream of the *start-EGFP-AID-3xHA-linker* cassette. The homology repair template was constructed in a modified pBluescript vector with a 3xP3>dsRed cassette in the vector backbone that could be used to screen out unwanted whole plasmid integrants produced by homologous recombination. The gRNA was expressed using the pU6 BbsI vector. Likely transformants were identified by measuring the frequency of mutations induced by *ebony* co-CRISPR as described previously (92). Individual edited chromosomes were identified using PCR and correct site integration was verified by Sanger sequencing.

Pho: Because prior reports indicated a lack of complementation for N-terminal GFP-Pho chimeras (39), we chose to target the single Pho C-terminus with a *linker-3xHA-AID-sfGFP-stop* cassette. Genome-edited *pho-sfGFP* are homozygous-viable. In experiments not presented in the current manuscript, Jabba-Trap knockdown of *pho-sfGFP* yields reduced fecundity of adult females and the majority of eggs fail to develop to the cellular blastoderm stages. The efficacy of Pho knockdown via the AID tag has not been evaluated.

Pho gRNA target: chr4:1172891-1172910 (+) 5'- CATATACCACAAACGGTACA (TGG) -3'

The gRNA cuts 22 bp upstream of the annotated *pho* stop codon.

The homology repair template contained 273 bp of left- and 282 bp of right-homology sequence flanking the gRNA cut site, and a silent G>A mutation at position 3 of the PAM sequence to prevent cutting of the repair construct. The homology repair template was constructed in the pJAT7 vector (93) which also drove expression of the gRNA. pJAT7 inserts were synthesized by Twist Biosciences and introduced with an LR Clonase reaction as described previously (93). To facilitate selection of positive insertions, the *sfGFP* sequence was split at a TTAA sequence and a PiggyBac cassette containing a 3xP3>dsRed cassette was inserted, allowing for scoring of genome-edited individuals by red fluorescence in the eye. Whole plasmid integrants typically expressed substantially higher levels of 3xP3>dsRed fluorescence. Individual dsRed+transformants without whole plasmid integration were verified using TagMap (94) followed by Nanopore sequencing (Plasmidsaurus). Individuals homozygous for the C-terminal PiggyBac 3xP3>dsRed cassette died as pharate adults. The PiggyBac-dsRed cassette was excised by crossing single adult dsRed+ males to *alpha-tubulin>PB-transposase; Gat[eya]/Ci[D]* (product of crossing Bloomington Drosophila Stock Center stocks 32070 and 90852) and then by crossing the resulting dysgenic male progeny to *yw*;

Gat[eya]/Ci[D] virgins. F2 males from the dysgenic cross were screened for loss of the PiggyBac cassette and individual stocks were verified by Nanopore sequencing (Plasmidsaurus) of PCR products spanning the sfGFP insertion site.

GAF: An *EGFP-3xHA-linker* sequence was added to the N-terminus of the *Trl* locus (encoding GAF) by phi-C31 mediated insertion of a *attB-EGFP-3xHA-Trl_{exon1}-intron-LoxP-vector* sequence into a CRISPR modified stock, *attP-3xMyc-Trl_{exon1}-intron-LoxP*, kindly provided by Tsutomu Aoki and Paul Schedl (Princeton University). Insertion of the EGFP cassette into the target locus results in a duplication of the *Trl* N-terminus where the outer duplication contains the engineered EGFP cassette, and the inner duplication is the original N-terminus. The insertion provides a second LoxP site that allows for the subsequent excision of the original *Trl* N-terminus leaving only the *EGFP-3xHA-Trl* coding sequence with an upstream *attL* and a single residual intronic *LoxP* site. Excision of the duplicated *Trl* N-terminal sequence was performed by crossing transformants to *y w; MKRS/TM6B p{mini-white = Crew}DH2* (Bloomington Drosophila Stock Center 1501). Genome-edited *EGFP-Trl* individuals are homozygous-viable. Sequences were verified by Sanger sequencing of PCR products flanking the modified locus.

Zelda Germline Clones: As described previously (70), *zelda* germline clones were made by crossing *w zld[294] FRT19A/FM7h; RpA-70-EGFP* females to *ovoD^{hs>flp} FRT19A/Y* males and performing 3x 1- hour 37°C heat shocks on days 3-5 of development.

Jabba-Trap: Based on the original reported design for Jabba-Trap (56), we designed a transgene that would yield maternal expression of Jabba-Trap directly driven by the maternal *alpha-Tubulin 67C* regulatory region. To do this, we synthesized the cDNA encoding the RG isoform of *jabba* chimerically fused to an anti-GFP nanobody (95) on both the N- and C- terminus, introducing 30 bp of codon optimization to the 5' end of the N-terminal and 3' end of the C-terminal anti-GFP nanobodies to allow for PCR amplification. Synthesized DNA was amplified by primers that appended NotI and NheI sites to the construct, and this product was cloned into a transgenic vector (pBARR) containing the 5' regulatory sequences for *alpha-tubulin 67C* through the first intron and the *spaghetti squash* 3' UTR. pBARR contains an *attB* site suitable for phi-C31 transgenesis. Transgenic lines were established at the attP40 and VK000033 attP landing sites by microinjection of the transgenic vector into phiC-31 expressing embryos.

The JT-GAF knockdown experiment was performed in embryos derived from the cross between *tub67C>JabbaTrap{attP40}/+; EGFP-Trl* mothers and *EGFP-Trl* fathers yielding embryos where both maternal and zygotic EGFP-GAF protein is sequestered by maternally expressed Jabba-Trap. Phenotypic details of this knockdown beyond effects on H3K27me3 will be reported elsewhere.

Additional stocks used

- *y w; {mini-white = RpA-70-EGFP}attP2* (70)
- *w p{mini-white = His2Av-RFP}1* (70)
- *w zld[294] FRT19A/FM7h* (24)
- *w p{mini-white = His2Av-RFP}1 zld[294] FRT19A/FM7h* (recombinant of the two prior stocks)
- *C(1)DX y w f/ p{mini-white = ovoD1-18}P4.1, p{ry = hs>Flp}12, y w sn FRT19A* (70)
- *E(z)[61]/TM3, Sb* (96)

Plasmids

Vectors for performing CRISPR mutagenesis are as follows:

- [pBS-3xP3>RFP](#) (92 [DOI](#)). Cloning of HDR constructs for classic CRISPR approaches.
- pU6-2-BbsI-gRNA (gift from Kate O'Connor-Giles, Drosophila Genomics Resource Center).

Cloning of gRNA constructs for classic CRISPR approaches.

- [pJAT7](#) (gift from David Stern (Addgene plasmid # 204292)). Cloning of gRNA and HDR sequences for new CRISPR approaches.

ChIP Sample Preparation

Timed collections of the indicated genotypes were performed, and embryos were dechorionated for 1-2 minutes in 4% sodium hypochlorite followed by extensive washing in deionized water. Dechorionated embryos were crosslinked by incubation for 15 minutes in a biphasic solution of 1x Phosphate Buffered Saline with 0.5% Triton X-100 : Heptane :: 2 ml : 6 ml to which 180 μ l of 20% paraformaldehyde was added. Crosslinked embryos were collected by centrifugation and the crosslinking reaction was stopped by a 1-minute incubation in 1x PBS + 0.5% Triton X-100 + 125 mM Glycine followed by three washes in 1x PBS + 0.5% Triton X-100. Crosslinked embryos were sorted by transferring the sample to a 35 mm petri dish lined with 1% agarose overlaid with a large volume of 1x PBS + 0.5% Triton X-100. For samples sorted by cell cycle (NC13, NC14E/M/L), the maternal genotype included RpA-70-EGFP which produces bright nuclear fluorescence scorable under an epifluorescent stereoscope at 100x magnification. NC14 embryos were subdivided into Early/Mid/Late groups by measuring the degree of nuclear elongation and cellularization as described previously (70 [DOI](#)). For measurement of E(z) binding comparing wild type and *zelda* germline clones, embryos were sorted using His2Av-RFP expression and the maternal genotypes for this experiment were: wild type = *w* *p*{His2Av-RFP}1; *EGFP-E(z)*/⁺ and *zelda* (clones) = *w* *p*{His2Av-RFP}1 *zld*[294] *FRT19A*; *EGFP-E(z)*/⁺. To stage *zelda* mutant collections, we relied on the following criteria. Embryos were sorted first based on nuclear density, selecting NC14 embryos with regularly spaced nuclei (i.e., prior to significant nuclear clustering and fallout typical of late NC14 *zelda* mutants). To distinguish early and mid NC14 embryos, we relied on the severity of the *halo* mutant phenotype typical of embryos with inhibited zygotic genome activation. Sorted embryos were collected in 1.5 ml tubes, excess media was removed, and samples were frozen at -80°C until needed.

For a particular experiment, a single biological replicate consisted of a 10-embryo equivalent of sheared chromatin produced from a batch of staged crosslinked embryos. Compared with prior ChIP-seq approaches (e.g., (70 [DOI](#))), we amended as follows our sample preparation protocol to significantly increase the sensitivity of the approach by preparing ChIPped DNA ends by exposure to Tn5 transposase (tagmentation) prior to the typical wash steps of an immunoprecipitation. Aliquots of staged crosslinked embryos were resuspended in 700 μ l of RIPA buffer (Sigma R0278) supplemented with 1x protease inhibitor cocktail (Sigma P8340) and 1 mM Dithiothreitol. Chromatin was partly sheared by performing four 15-second rounds of sonication with a Branson 450 sonicator equipped with a 1/4-inch microtip horn (20% output, full duty cycle). 10-embryo equivalents of sheared chromatin were distributed to 1.5 ml DNA Lo-Bind tubes (Eppendorf) and brought to 650 μ l final volume with RIPA buffer supplemented with protease inhibitors. Primary antibodies were added to the sheared chromatin preparations and incubated overnight at 4°C on a nutator. For each planned sequencing run, several biological replicates of a negative control ChIP were performed. Two negative control strategies were employed for this study. In most cases, an aliquot of sheared chromatin was immunoprecipitated with a rabbit polyclonal antibody to the c-myc epitope tag, which is absent from any of the genotypes in this study. Alternatively, for experiments measuring enrichment of GFP-tagged chimeric proteins, a rabbit polyclonal antibody to GFP was used as a negative control on wild type (*w*[1118]) chromatin. Protein G Dynabeads (15 μ l per ChIP, Thermo-Fisher) were blocked on a nutator overnight in a solution of 1x PBS, 0.1% Triton X-100, 3% Bovine Serum Albumin (Sigma), and 0.2 mg/ml Yeast tRNA (Roche). The following morning, 15 μ l of blocked Dynabeads were added to the sheared chromatin samples and

incubated for one hour. Dynabeads were then immobilized on a magnetic stand and the captured material was exchanged into Tagmentation Wash Buffer (10 mM Tris pH 7.5, 5 mM MgCl₂) by performing two 1 ml washes. Samples were then subjected to tagmentation by resuspending the Dynabead complexes in 20 µl of 1x Tn5 reaction buffer containing 1 µl of Tn5 transposase (Illumina Nextera) and incubating for 40 minutes in an Eppendorf Thermomixer set at 37°C with 1000 RPM shaking. Next, Dynabead complexes were subjected to five-minute washes, twice in “Wash buffer I” (20 mM Tris pH8.0, 0.1% SDS, 1% Triton X-100, 2mM EDTA, 150 mM NaCl), twice in “Wash buffer II” (Wash buffer I with 500 mM NaCl), and once in TE buffer (10 mM Tris pH 8.0, 1 mM EDTA). Tagmented, immunoprecipitated DNA fragments were eluted from the Dynabeads and were crosslink-reversed by incubation in “Elution Buffer” (50 mM Tris pH8.0, 10 mM EDTA, 1% SDS) containing 100 µg of Proteinase K (New England Biolabs) for four hours in an Eppendorf Thermomixer set at 65°C with 1500 RPM shaking. The eluted DNA fragments were purified using a Qiagen MinElute Enzyme Reaction Cleanup kit, performing one extra PE wash for three minutes and eluting after a three- minute incubation in 22.5 µl Qiagen Buffer EB.

ChIP-seq Library Preparation and Sequencing

Following purification of tagmented ChIP samples, library preparation was performed by setting up PCRs containing 22.5 µl eluted ChIP DNA, 25 µl NEBNext Q5 2x PCR Mastermix (New England Biolabs), and

2.5 µl of a unique-dual-indexed primer mix (25 µM each primer). Primer designs were as previously reported ([92](#)). Before performing any thermal cycling, reactions were mixed well and centrifuged, and a 5 µl aliquot of the primary PCR solution (pre-amplification) was added to a second, “side-reaction” that consisted of 5 µl primary PCR solution, 5 µl NEBNext Q5 2x PCR Mastermix, and 5 µl of a solution of

1.25 µM unique-dual-indexed primer mix in 1.8x SYBR-Green I (Thermo-Fisher). The primary PCR samples were stored at 4°C while the side-reaction was performed.

The side-reaction was subjected to QPCR to determine the total number of cycles necessary to amplify the primary PCR using the following cycling parameters: 1x 72°C, 5 minutes; 1x 98°C, 30 seconds; 20x (98°C, 10 seconds; 63°C, 30 seconds; 72°C 30 seconds). The total number of cycles to apply to the primary PCR was determined by finding the side-reaction cycle number (*N*) necessary to yield QPCR fluorescence intensity equal to 1/3 of the plateau fluorescence intensity. Based on these measurements, the primary PCR samples were then amplified using the following cycling parameters: 1x 72°C, 5 minutes; 1x 98°C, 30 seconds; *N*x (98°C, 10 seconds; 63°C, 30 seconds; 72°C 30 seconds); 1x 72°C, 5 minutes.

Barcoded DNA from the primary PCRs was purified by 1.8x Ampure, following the manufacturer’s instructions. DNA concentrations were determined by Qubit fluorometry using 1x HS DNA reagents (Invitrogen). The size-distribution of libraries was determined by TapeStation analysis using D5000 screen tapes (Agilent).

Up to 48 libraries were pooled together in equimolar concentrations and submitted to Admera Health (South Plainfield, NJ) for paired-end (2x 150 bp) sequencing using an Illumina NovaSeqX platform. One early dataset, RNA Pol2 CTD-pSer5 ChIP in wild type versus *zelda* was sequenced on a NextSeq500 platform at Northwestern University.

ChIP-seq data processing and read mapping

ChIP-seq data were adaptor-trimmed and mapped on Northwestern University’s QUEST high-performance computing cluster. Demultiplexed fastq files were subjected to adaptor-trimming using TrimGalore v.0.6.10 (<https://github.com/FelixKrueger/TrimGalore>) and cutadapt v4.4 ([97](#)) and mapped to the *Drosophila melanogaster* dm6 reference genome assembly using bowtie2 v2.4.1

(98 [↗](#)) with option -X 2000. Suspected optical and PCR duplicates were marked by Picard MarkDuplicates version 2.21.4 (<https://broadinstitute.github.io/picard/> [↗](#)) using default parameters. Trimmed, mapped, and duplicate- marked reads were imported into R using the GenomicAlignments (version 1.38.2, (99 [↗](#))) and Rsamtools (version 2.18.0, (100 [↗](#))) packages. Properly paired reads with primary mappings with map-quality scores greater than or equal to 10 were imported. Only reads corresponding to chromosome-level assemblies (chr2L, 2R, 3L, 3R, X) were considered further. Coverage plots (wiggle plots) were generated using the GenomicRanges package (version 1.42.0, (99 [↗](#))).

Peak calling: Regions of enrichment for transcription factor ChIP targets were called using MACS2 v2.2.7.1 (101 [↗](#)) on merged replicate samples, using a GFP ChIP in wild type embryos as a negative control. In practice, peak calling results from initial adjusted p-value thresholds of $q = 1e-5$ were visually inspected against coverage plots for the ChIP target and negative-control ChIP to determine whether the selected adjusted p-value threshold adequately captured specific enrichment of the target factor. If $q = 1e-5$ was found to produce less than optimal results, the threshold was changed and the process repeated until the peak calling results were deemed by the user to accurately reflect biologically meaningful enrichments. The following are the final chosen q-value thresholds: E(z), $1e-7$; Pho, $1e-40$; Cg, $1e-30$; GAF, $1e-5$. Zelda peaks were generated by converting reported peaks (33 [↗](#)) from the dm3 to the dm6 reference genome using UCSC LiftOver in the rtracklayer package (version 1.50.0, (102 [↗](#))).

Broad regions of enrichment for H3K27me3 ChIP experiments were determined as described previously (35 [↗](#)) using the STAN package in R (version 2.18.0, (103 [↗](#))). H3K27me3 ChIP-seq coverage and negative control ChIP-seq coverage from late NC14 wild type samples were used for classifying genomic regions into three states (low me3/low background, high me3/low background, and high background). The high me3/low background state was processed further. To merge contiguous tracts of high H3K27me3 signal, genomic regions in the high me3/low background state were widened by 20 kbp, overlapping regions were merged, and 20 kb was subtracted from the resulting merged regions. Merged regions less than 4 kb in width were dropped from the list. This yielded a mapping of provisional Polycomb Domains. Following visual examination of the provisional mappings, a small number of regions corresponding to low-magnitude enrichment of H3K27me3 ChIP-seq coverage were dropped to simplify downstream analysis.

Differential Enrichment Analysis

The procedure for performing differential enrichment analysis for H3K27me3 between wild type, Jabba- trap GAF and *zelda* germline clone embryos is presented in detail in the supplemental information. Briefly, the dm6 reference genome assembly was subdivided into 2 kb bins spanning the canonical chromosome assemblies and the number of reads in each mapped ChIP sample overlapping with each bin was tabulated. Bins with low read counts were filtered out and differential enrichment between genotypes was calculated using DESeq2 (version 1.30.1, (57 [↗](#))) using formulae with the form $\sim batch + genotype$, where “batch” reflected any batch effects identified through a preliminary principal component analysis. To identify consecutive runs of bins with statistically significant changes in ChIP signal, the DESeq results including adjusted p-values and log2fold changes were appended to a GenomicRanges object representing the genomic coordinates of each associated bin, and bins with adjusted p-values greater than 0.05 were filtered out. The remaining bins with adjusted p-values less than 0.05 were widened on each end by one bp to force overlaps with adjoining significant bins. Overlapping widened bins were then merged and one bp was removed from the ends of the remaining ranges after merging, yielding a mixture of new runs of consecutive significant bins and singleton bins. To propagate log2 fold-change values,

the average log₂ fold change of the bins comprising a run was calculated and appended to the run's metadata. To combine p-value information, we applied Fisher's method by calculating for n p-values (p) associated with bins in a run the right-tail χ -squared p-value for

$$-2 \sum_{i=1}^n \log(p_i)$$

with $2n$ degrees of freedom (104 [link](#)).

Live Imaging

Embryos expressing an EGFP chimera (E(z), Pho, Cg, or GAF) and His2Av-RFP were collected from 0-2 hour plates and dechorionated in 4% sodium hypochlorite for 1 minute followed by extensive washing in deionized water. Dechorionated embryos were mounted in a small drop of Halocarbon 27 oil (Sigma- Aldrich) between a size 1.5 coverslip and a gas-permeable membrane (Biofoil, Heraeus) stretched over an acrylic holder.

Mounted samples were imaged on a Leica SP8 laser scanning confocal microscope equipped with a white light laser using a 63x/1.3 NA glycerol-immersion objective. 512 x 256 pixel images were captured at 2x zoom with 400 Hz scan rate. Z-stacks were collected over the entire nuclear volume with a step size of 1 μ m. Frames were collected for approximately two hours imaging every 30 seconds from metaphase of nuclear cycle 10 through one hour past metaphase 13 (approximately the start of gastrulation). Laser power was chosen to maximize intensity and minimize photobleaching in the GFP channel, as estimated by taking a zoomed-out overview image of the embryo following completion of the imaging session and verifying that bleaching was minimal. It was permissible for the His2Av-RFP imaging conditions to yield bleaching since this channel was only used for nuclear segmentation and not quantified.

To quantify GFP nuclear intensity, each time frame of a 3-dimensional image stack was segmented in the Histone RFP channel by first thresholding the image using Otsu's method, filling in border pixels by dilation followed by erosion, and then breaking conjoined nuclei by using a watershed transformation. This yielded a nuclear mask from which the average pixel intensity in the GFP channel was calculated for each individual nucleus. This procedure was repeated over all frames of the movie. The average nuclear intensity for all nuclei in a frame was calculated, yielding the raw signal intensity for the movie. The raw signal intensity was next normalized to the mean intensity of the entire movie. To account for biological differences in nuclear cycle length between replicates, frames within individual nuclear cycles were binned to yield standard nuclear cycle durations of NC11 = 10 minutes, NC12 = 12.5 minutes, NC13 = 20 minutes, and NC14 = 60 minutes. Following binning for nuclear cycle length, average normalized intensities across replicates was calculated and plotted.

Immunostaining

Immunostaining was performed on embryos fixed either by heat-methanol (H3K27me3) or formaldehyde (H2Aub, E(z), Pho, Cg, GAF) approaches. For either approach, embryos from 0-4 hour collections were first dechorionated in 4% Sodium Hypochlorite for 1-2 minutes followed by extensive washing in deionized water. Heat-Methanol fixation was performed by plunging dechorionated embryos into a 20 ml scintillation vial containing 5 ml of boiling 0.3% Triton X-100 + 0.4% (w/v) NaCl for ten seconds before placing on ice to cool rapidly. After cooling, the solution was removed and replaced with 5 ml heptane and 5 ml methanol to 'pop' embryos from their vitelline membranes. Formaldehyde fixation was performed by adding dechorionated embryos to a biphasic solution of 5 ml Heptane: 4 ml 1x phosphate buffered saline containing 1 ml of 20% paraformaldehyde solution in a 20 ml scintillation vial. The solution was placed on an orbital shaker for 20 minutes with 250 RPM shaking. The lower aqueous phase was carefully removed and replaced with 5 ml of methanol to "pop" embryos from their vitelline membranes. For both

fixation methods, popped embryos were collected from the bottom of the scintillation vial and washed 3x in methanol prior to freezing for ≥ 24 hours in methanol. In preliminary trials for a particular antigen target, both fixation methods were performed, and a particular approach was chosen based on how well an antigen target would be stained under each fixation condition. For reasons that are not clear, formaldehyde fixation yielded poor signal even in late-stage embryos for H3K27me3 stainings, with excessive cytoplasmic background. Heat-methanol fixation greatly reduced Cg signals and caused excessive background with the H2Aub staining. Formaldehyde fixation was chosen for E(z), Pho, Cg, and GAF to maintain similar fixation conditions for these four factors which were to be compared directly.

The immunostaining procedure involved rehydration of fixed embryos in PTx (1x phosphate buffered saline + 0.1% Triton X-100), followed by one hour of blocking in PTx + 5% Bovine Serum Albumin. The primary antibody incubation was performed in blocking buffer overnight at 4°C on a nutator. Anti GFP, H2Aub, and H3K27me3 were used at a 1:500 dilution. Following the primary incubation, samples were washed for 40 minutes in four changes of PTx + 1% Bovine Serum Albumin. Secondary antibody incubations were performed in blocking buffer for two hours at room temperature on a nutator. The secondary antibody (goat anti rabbit coupled to alexa 546) was used at a 1:500 final dilution after pre-adsorption against fixed embryos at a 1:10 dilution. DAPI was added to 0.1 mg/ml final concentration during the final hour of secondary antibody incubation to stain DNA. If Alexa 647-coupled WGA (Invitrogen W32466) was included in a staining, it was added at a 1:100 final dilution at the same time as the DAPI treatment. Following the secondary incubation, samples were washed for 40 minutes in four changes of PTw (1x phosphate buffered saline + 0.1% Tween-20) prior to mounting embryos on slides in ProLong Gold (Invitrogen) with a size 1.5 coverslip, sealing prior to curing with clear nail polish.

Mounted samples were imaged on a Leica SP8 laser scanning confocal microscope equipped with a white light laser using a 63x/1.3 NA glycerol-immersion objective. 512x512 pixel images were captured at 2x zoom with 100 Hz scan rate. Z-stacks were collected over the entire nuclear volume with a step size of 1 μ m. Alexa 546 conjugated secondary antibodies were excited at 557 nm. WGA coupled to Alexa 647, if used, was excited at 653 nm. DAPI was excited using a 405 nm laser.

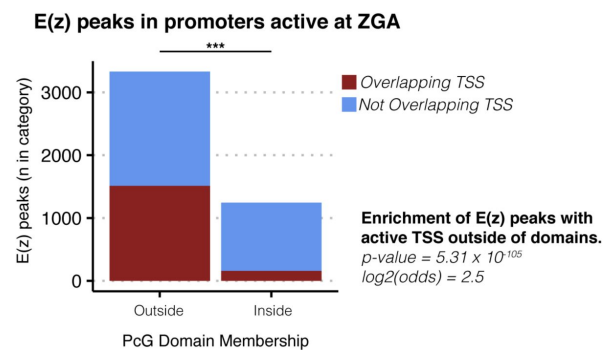
Antibodies

Antibodies were purchased from commercial sources and used as listed below.

Antibody	Source	Identifier	Technique
H3K27me3 Antibody	Diagenode	C15410195	ChIP-seq, IF
Anti-Green Fluorescent Protein Antibody	Millipore Sigma	AB3080	ChIP-seq, IF
Ubiquityl-Histone H2A (Lys119) (D27C4)	Cell Signaling	8240	ChIP-seq, IF
MonoMethyl-Histone H3-K27 Rabbit pAb	Abclonal	A2361	ChIP-seq
Anti-RNA polymerase II CTD repeat YSPTSPS (phospho S5)	abcam	ab5131	ChIP-seq
Mono-Methyl-Histone H3 (Lys4) (D1A9)	Cell signaling	5326T	ChIP-seq
H3K4me2 Recombinant Superclonal Antibody, ChIP-Verified	Fisher	710796	ChIP-seq
TriMethyl-Histone H3-K4 Rabbit pAb	Abclonal	A2357	ChIP-seq
H3K27ac Monoclonal Antibody, ChIP-Verified	Invitrogen	MA5-23516	ChIP-seq
Alexa Fluor 546 Goat anti Rabbit	Invitrogen	A11035	IF

The anti H3K27me1 and me3 antisera chosen for this study have previously been subjected to specificity testing using the methodology reported here (105 [↗](#)). Specificity data were recovered from <https://www.chromatinantibodies.com> [↗](#).

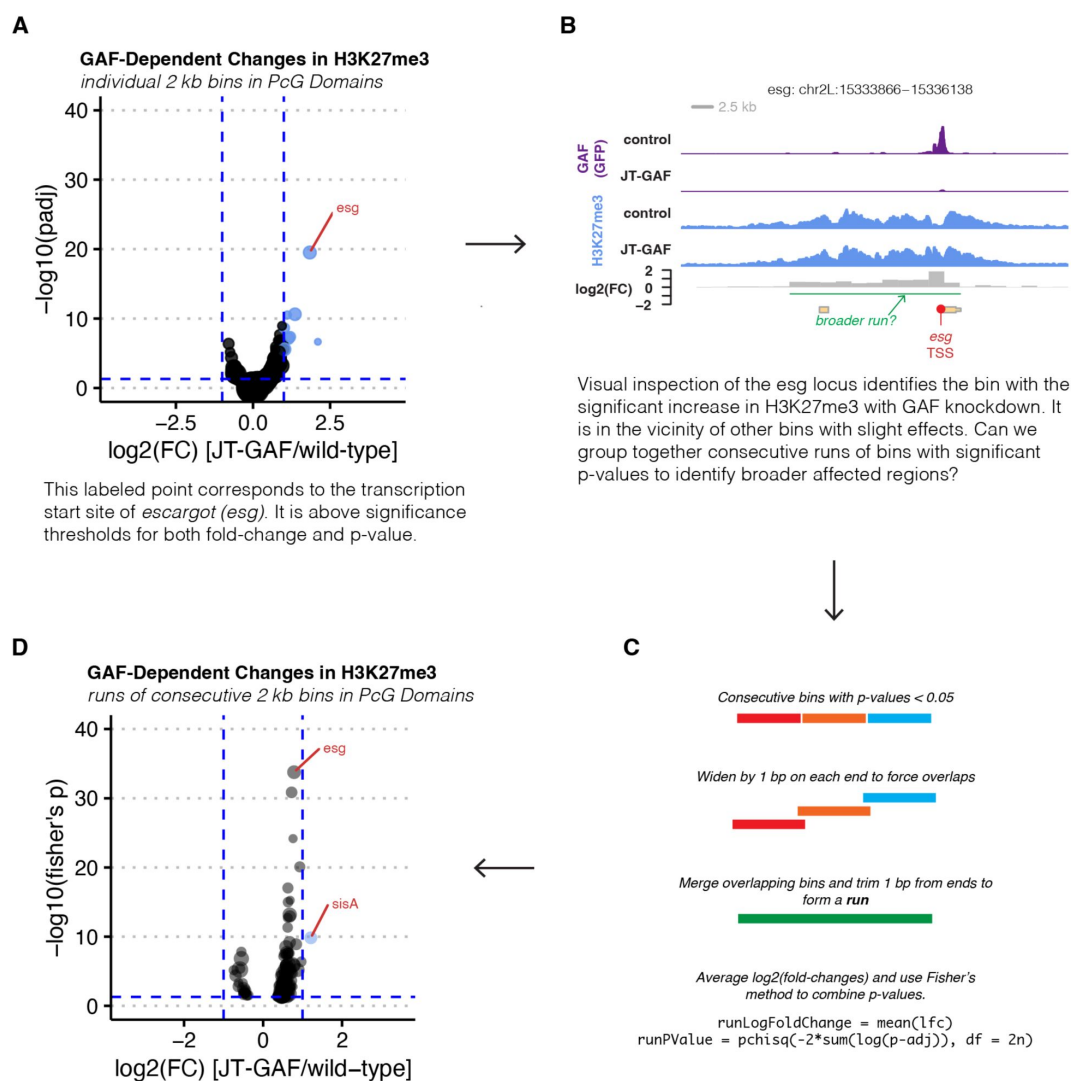
Supplementary figures



Supplemental Figure 1

The number of E(z) peaks inside and outside of PcG Domains are shown in the bar graph, subset by whether a peak directly overlaps with an annotated transcription start site.

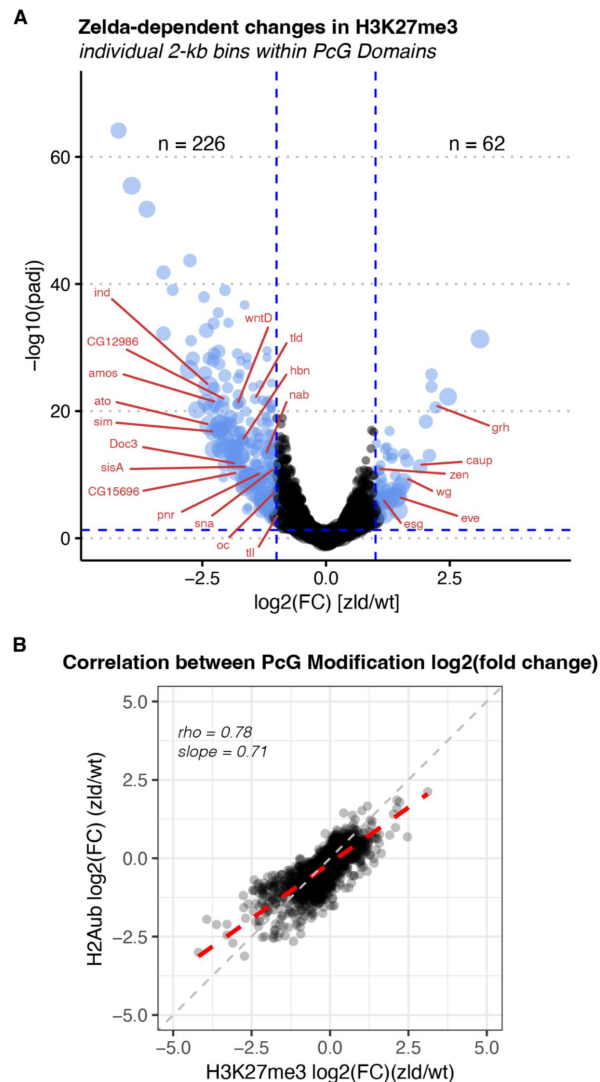
We note that in comparison to the heatmaps this overlap measurement likely under-estimates the degree of overlap outside of PcG domains. This could be due to under-counting of active TSS, and/or a peak not strictly overlapping a TSS, but instead being near-by. Nevertheless, E(z) peaks outside of domains are significantly enriched for annotated active TSS regions with a log2 odds ratio of 2.5 and a Fisher's exact p-value of 5.31×10^{-105} .



Supplemental Figure 3

DESeq on 2 kb bins for the GAF knockdown experiment and making runs out of bins.

A) A volcano plot is shown for DESeq2 analysis of H3K27me3 ChIP-seq for EGFP-GAF versus JT-GAF. Overall, very few 2 kb bins show effects that surpass the significance thresholds for p-value and $\log_2(\text{fold change})$. The bin corresponding to the *escargot* transcription start site is labeled. B) Reproduced from [Figure 6](#) with additional annotations indicating the location of the highlighted point in panel A and a broader region surrounding the bin with a more moderate effect on H3K27me3 following JT-sequestration of GAF. C) Strategy for generating consecutive 2 kb runs of bins with p-values < 0.05, and how fold-change and significance values are propagated. D) Reproduced from [Figure 6](#) to show how making consecutive runs transforms the data.



Supplemental Figure 4

DESeq on 2 kb bins for the *zelda* mutant experiment and correlation between H3K27me3 and H2Aub.

A) A volcano plot is shown for DESeq analysis performed on H3K27me3 ChIP seq between *zelda* mutants and wild type embryos, staged mid NC14. 226 bins have significant reductions in H3K27me3, and 62 bins have significant increases. Significant bins containing a zygotic transcription start site are labeled with the gene symbol (red). B) Log2(fold change) values from DESeq analysis of H3K27me3 (x-axis) and H2Aub (y-axis) ChIP-seq between *zelda* and wild type embryos are plotted. The correlation coefficient is 0.78. Linear regression was performed, and the fit model is plotted in red (slope = 0.71). There is a strong positive correlation between changes in both modifications following loss of *zelda*, although the trend is for H2Aub to lag slightly in its magnitude of response compared with H3K27me3.

Acknowledgements

We thank Eric Wieschaus for comments and critiques on an earlier version of this manuscript. We thank Paul Schedl and Tsutomu Aoki for generously providing the *attP-3xMyc-Trl-loxP* stock, and Yue Yang and Jason Brickner for sharing access to instrumentation for sequencing and library preparation. We thank the Bloomington Drosophila Stock Center for providing stocks and Flybase for providing an essential resource to the *Drosophila* community. N.G-S., E.A.D., and C.C. were supported by the Cellular and Molecular Basis of Disease training program (T32 GM008061). E.A.D. is supported by an NSF GRFP fellowship (DGE-2234667) and is a Data Science Fellow at the Northwestern Institute on Complex Systems. Experiments were supported by the National Institutes of Health grant R01 HD101563 to S.A.B.. S.A.B. is a Pew Scholar in the Biomedical Sciences, supported by the Pew Charitable Trusts.

Additional information

Data Availability

ChIP-seq datasets generated for this study have been deposited in Gene Expression Omnibus (accession number: [GSE299311](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE299311)). Files for reproducing the differential enrichment analysis and code can be accessed at https://github.com/sblythe/Gonzaga_Saavedra_2025_Code_Supplement 

Additional datasets used

The Zelda ChIP-seq track and peaks list are from the 3-hour timepoint from Harrison et al, 2011 ((33 ), GSE30757).

ATAC-seq tracks comparing NC14 wild type and *zelda* germline clone chromatin accessibility are from Hannon et al, 2017 ((59 ), GSE86966)

Author contributions

N.G-S.: Conceptualization (equal), Data Curation (equal), Formal Analysis (equal), Investigation (lead), Visualization (equal), Writing-Original Draft Preparation (supporting), Writing-Review & Editing (equal). **E.A.D.:** Conceptualization (supporting), Data Curation (supporting), Formal Analysis (supporting), Investigation (supporting), Visualization (supporting), Writing-Original Draft Preparation (supporting), Writing-Review & Editing (equal). **I.V.S.:** Investigation (supporting), Formal Analysis (supporting), Resources (supporting). **C.C.:** Conceptualization (supporting), Formal Analysis (supporting), Resources (supporting). **S.A.B.:** Conceptualization (equal), Data Curation (supporting), Formal Analysis (equal), Funding Acquisition (lead), Project Administration (lead), Visualization (equal), Writing-Original Draft Preparation (lead), Writing-Review & Editing (equal).

Funding

Eunice Kennedy Shriver National Institute of Child Health and Human Development (R01 HD101563)

Pew Charitable Trusts

National Institute of General Medical Sciences (T32 GM008061)

National Science Foundation (DGE-2234667)

Additional files

Code Supplement. [↗](#)

Supplemental Movie 1. His2Av-RFP; EGFP-E(z) NC10 to 60 minutes into NC14. A representative movie of (top) EGFP-E(z) and (bottom) a merged image with His2Av-RFP (magenta) and EGFP-E(z) (yellow) is shown. E(z) localizes to interphase nuclei throughout the period of NC10-NC14. [↗](#)

Supplemental Movie 2. His2Av-RFP; Pho-sfGFP NC10 to 60 minutes into NC14. A representative movie of (top) Pho-sfGFP and (bottom) a merged image with His2Av-RFP (magenta) and Pho-sfGFP (yellow) is shown. Pho is initially excluded from nuclei and increases in concentration between NC11 and NC14. [↗](#)

Supplemental Movie 3. His2Av-RFP; EGFP-Cg NC10 to 60 minutes into NC14. A representative movie of (top) EGFP-Cg and (bottom) a merged image with His2Av-RFP (magenta) and EGFP-Cg (yellow) is shown. Cg is initially excluded from nuclei and increases in concentration between NC12 and NC14. [↗](#)

Supplemental Movie 4. His2Av-RFP; EGFP-GAF NC10 to 60 minutes into NC14. A representative movie of (top) EGFP-GAF and (bottom) a merged image with His2Av-RFP (magenta) and EGFP-GAF (yellow) is shown. GAF initially only localizes to pericentric GA-rich regions before increasing nucleoplasmic concentration between NC11 and 14. [↗](#)

Supplemental Movie 5. His2Av-RFP; EGFP-GAF NC10 through NC14, control for Jabba-Trap knockdown experiment. This is a control movie to be compared with Supplemental Movie 6, imaged under identical conditions. The movie shows a single z-slice of an His2Av-RFP; EGFP-GAF embryo from NC10 to NC14, with single-channel EGFP-GAF at left (grey), and the merged image at right (green = EGFP-GAF). [↗](#)

Supplemental Movie 6. His2Av-RFP; Jabba-Trap/+; EGFP-GAF NC10 through NC14. This movie shows the effect of Jabba-Trap sequestration on EGFP-GAF in a single z-slice of an His2Av-RFP; Jabba-Trap/+; EGFP-GAF embryo, with single-channel EGFP-GAF at left (grey), and the merged image at right (green = EGFP-GAF). [↗](#)

Supplemental Data Table 1. Annotated Polycomb Domains. The list of PcG Domains used in the text is presented in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric), End (numeric), Width (numeric), Strand (character), Domain Name (character), with E(z) (logical), with Cg (logical), with GAF (logical), with Zld (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with Zygotic-only transcription start site (logical), Zygotic-only transcription start site gene name (character). [↗](#)

Supplemental Data Table 2. Annotated E(z) Peaks. The list of E(z) peaks used in the text is presented in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric), End (numeric), Width (numeric), Strand (character), MACS Signal Value (numeric), MACS q-value (numeric), peak summit (numeric), within a PcG Domain (logical), PcG Domain Name (character), Maximal E(z) peak in Domain (logical), with Pho (logical), with Cg (logical), with GAF (logical), with Zld (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with a Zygotic-only transcription start site (logical), Zygotic-only transcription start site name (character). [↗](#)

Supplemental Data Table 3. Annotated Pho Peaks. The list of Pho peaks used in the text is presented in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric),

End (numeric), Width (numeric), Strand (character), MACS Signal Value (numeric), MACS q-value (numeric), peak summit (numeric), within a PcG domain (logical), PcG Domain Name (logical), with E(z) (logical), with Cg (logical), with GAF (logical), with Zld (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with a Zygotic-only transcription start site (logical), Zygotic-only transcription start site name (character). [↗](#)

Supplemental Data Table 4. Annotated Cg Peaks. The list of Cg peaks used in the text is presented in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric), End (numeric), Width (numeric), Strand (character), MACS Signal Value (numeric), MACS q-value (numeric), peak summit (numeric), within a PcG domain (logical), PcG Domain Name (logical), with E(z) (logical), with Pho (logical), with GAF (logical), with Zld (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with a Zygotic-only transcription start site (logical), Zygotic-only transcription start site name (character). [↗](#)

Supplemental Data Table 5. Annotated GAF Peaks. The list of GAF peaks used in the text is presented in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric), End (numeric), Width (numeric), Strand (character), MACS Signal Value (numeric), MACS q-value (numeric), peak summit (numeric), within a PcG domain (logical), PcG Domain Name (logical), with E(z) (logical), with Pho (logical), with Cg (logical), with Zld (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with a Zygotic-only transcription start site (logical), Zygotic-only transcription start site name (character). [↗](#)

Supplemental Data Table 6. Annotated Zelda Peaks. The list of 3h after egg laying Zelda peaks from (Harrison et al, 2011) was lifted-over to dm6 coordinates. Here we report this list in tab-delimited format with additional annotations indicating co-occurrence with genomic features of interest. The columns are: Row number (numeric), Seqnames (Chromosome, character), Start (numeric), End (numeric), Width (numeric), Strand (character), Name (character), Peak Score (numeric), within a PcG domain (logical), PcG Domain Name (logical), with E(z) (logical), with Pho (logical), with Cg (logical), with GAF (logical), with a Maternal/Zygotic transcription start site (logical), Maternal/Zygotic transcription start site gene name (character), with a Zygotic-only transcription start site (logical), Zygotic-only transcription start site name (character). [↗](#)

References

1. Schuettengruber B, Bourbon HM, Di Croce L, Cavalli G (2017) **Genome Regulation by Polycomb and Trithorax: 70 Years and Counting** *Cell* **171**:34–57 [Google Scholar](#)
2. Laugesen A, Højfeldt JW, Helin K (2019) **Molecular mechanisms directing PRC2 recruitment and H3K27 methylation** *Mol Cell* **74**:8–18 [Google Scholar](#)
3. Müller J, Hart CM, Francis NJ, Vargas ML, Sengupta A, Wild B, et al. (2002) **Histone methyltransferase activity of a Drosophila Polycomb group repressor complex** *Cell* **111**:197–208 [Google Scholar](#)
4. Shao Z, Raible F, Mollaaghababa R, Guyon JR, Wu CT, Bender W, et al. (1999) **Stabilization of chromatin structure by PRC1, a Polycomb complex** *Cell* **98**:37–46 [Google Scholar](#)
5. Cao R, Wang L, Wang H, Xia L, Erdjument-Bromage H, Tempst P, et al. (2002) **Role of histone H3 lysine 27 methylation in Polycomb-group silencing** *Science* :298 [Google Scholar](#)
6. Czermin B, Melfi R, McCabe D, Seitz V, Imhof A, Pirrotta V (2002) **Drosophila enhancer of Zeste/ESC complexes have a histone H3 methyltransferase activity that marks chromosomal Polycomb sites** *Cell* **111**:185–96 [Google Scholar](#)
7. Margueron R, Justin N, Ohno K, Sharpe ML, Son J, Drury WJ, et al. (2009) **Role of the polycomb protein EED in the propagation of repressive histone marks** *Nature* :461 [Google Scholar](#)
8. Lee CH, Holder M, Grau D, Saldaña-Meyer R, Yu JR, Ganai RA, et al. (2018) **Distinct Stimulatory Mechanisms Regulate the Catalytic Activity of Polycomb Repressive Complex 2** *Mol Cell* **70**:435–448 [Google Scholar](#)
9. Lee CH, Yu JR, Kumar S, Jin Y, LeRoy G, Bhanu N, et al. (2018) **Allosteric activation dictates PRC2 activity independent of its recruitment to chromatin** *Mol Cell* **70**:422–434 [Google Scholar](#)
10. Kalb R, Latwiel S, Baymaz HI, Jansen PWTC, Müller CW, Vermeulen M, et al. (2014) **Histone H2A monoubiquitination promotes histone H3 methylation in Polycomb repression** *Nat Struct Mol Biol* **21**:569–71 [Google Scholar](#)
11. Kasinath V, Beck C, Sauer P, Poepsel S, Kosmatka J, Faini M, et al. (2021) **JARID2 and AEBP2 regulate PRC2 in the presence of H2AK119ub1 and other histone modifications** *Science* **371**:eabc3393 [Google Scholar](#)
12. Hagstrom K, Muller M, Schedl P (1997) **A Polycomb and GAGA dependent silencer adjoins the Fab-7 boundary in the Drosophila bithorax complex** *Genetics* **146**:1365–80 [Google Scholar](#)
13. Brown JL, Mucci D, Whiteley M, Dirksen ML, Kassis JA (1998) **The Drosophila Polycomb group gene pleiohomeotic encodes a DNA binding protein with homology to the transcription factor YY1** *Mol Cell* **1**:1057–64 [Google Scholar](#)

14. Horard B, Tatout C, Poux S, Pirrotta V (2000) **Structure of a polycomb response element and in vitro binding of polycomb group complexes containing GAGA factor** *Mol Cell Biol* **20**:3187–97 [Google Scholar](#)
15. Brown JL, Fritsch C, Mueller J, Kassis JA (2003) **The Drosophila pho-like gene encodes a YY1-related DNA binding protein that is redundant with pleiohomeotic in homeotic gene silencing** *Development* **130**:285–94 [Google Scholar](#)
16. Wang L, Brown JL, Cao R, Zhang Y, Kassis JA, Jones RS (2004) **Hierarchical recruitment of polycomb group silencing complexes** *Mol Cell* **14**:637–46 [Google Scholar](#)
17. Klymenko T, Papp B, Fischle W, Köcher T, Schelder M, Fritsch C, et al. (2006) **A Polycomb group protein complex with sequence-specific DNA-binding and selective methyl-lysine-binding activities** *Genes Dev* **20**:1110–22 [Google Scholar](#)
18. Ray P, De S, Mitra A, Bezstarosti K, Demmers JAA, Pfeifer K, et al. (2016) **Combgap contributes to recruitment of Polycomb group proteins in Drosophila** *Proc Natl Acad Sci U S A* **113**:3826–31 [Google Scholar](#)
19. Brown JL, Sun MA, Kassis JA (2018) **Global changes of H3K27me3 domains and Polycomb group protein distribution in the absence of recruiters Spps or Pho** *Proceedings of the National Academy of Sciences* **115**:E1839–48 [Google Scholar](#)
20. Coleman RT, Struhl G (2017) **Causal role for inheritance of H3K27me3 in maintaining the OFF state of a Drosophila HOX gene** *Science* **356**:eaai8236 [Google Scholar](#)
21. Laprell F, Finkl K, Müller J (2017) **Propagation of Polycomb-repressed chromatin requires sequence-specific recruitment to DNA** *Science* :356 [Google Scholar](#)
22. Foe VE, Alberts BM (1983) **Studies of nuclear and cytoplasmic behaviour during the five mitotic cycles that precede gastrulation in Drosophila embryogenesis** *J Cell Sci* **61**:31–70 [Google Scholar](#)
23. Blythe SA, Wieschaus EF (2016) **Establishment and maintenance of heritable chromatin structure during early Drosophila embryogenesis** *eLife* **5**:e20148 <https://doi.org/10.7554/eLife.20148> | [Google Scholar](#)
24. Liang HL, Nien CY, Liu HY, Metzstein MM, Kirov N, Rushlow C (2008) **The zinc-finger protein Zelda is a key activator of the early zygotic genome in Drosophila** *Nature* :456 [Google Scholar](#)
25. Gaskill MM, Gibson TJ, Larson ED, Harrison MM (2021) **GAF is essential for zygotic genome activation and chromatin accessibility in the early Drosophila embryo** *eLife* **10**:e66668 <https://doi.org/10.7554/eLife.66668> | [Google Scholar](#)
26. Duan J, Rieder L, Colonna MM, Huang A, McKenney M, Watters S, et al. (2021) **CLAMP and Zelda function together to promote Drosophila zygotic genome activation** *eLife* **10**:e69937 <https://doi.org/10.7554/eLife.69937> | [Google Scholar](#)
27. Harrison MM, Marsh AJ, Rushlow CA (2023) **Setting the stage for development: the maternal-to-zygotic transition in Drosophila** *Genetics* **225**:iyad142 [Google Scholar](#)
28. Li XY, Harrison MM, Villalta JE, Kaplan T, Eisen MB (2014) **Establishment of regions of genomic activity during the Drosophila maternal to zygotic transition** *eLife* **3**:e03737 <https://doi.org>

29. Reinig J, Ruge F, Howard M, Ringrose L (2020) **A theoretical model of Polycomb/Trithorax action unites sable epigenetic memory and dynamic regulation** *Nat Commun* **11**:4782 [Google Scholar](#)
30. Chen K, Johnston J, Shao W, Meier S, Staber C, Zeitlinger J. (2013) **A global change in RNA polymerase II pausing during the Drosophila midblastula transition** *eLife* **2**:e00861 [Google Scholar](#)
31. Zenk F, Loeser E, Schiavo R, Kilpert F, Bogdanović O, Iovino N (2017) **Germ line-inherited H3K27me3 restricts enhancer function during maternal-to-zygotic transition** *Science* **357** [Google Scholar](#)
32. Cardamone F, Piva A, Löser E, Eichenberger B, Romero-Mulero MC, Zenk F, et al. (2025) **Chromatin landscape at cis-regulatory elements orchestrates cell fate decisions in early embryogenesis** *Nat Commun* **16**:3007 [Google Scholar](#)
33. Harrison MM, Li XY, Kaplan T, Botchan MR, Eisen MB (2011) **Zelda binding in the early Drosophila melanogaster embryo marks regions subsequently activated at the maternal-to-zygotic transition** *PLoS Genet* **7**:e1002266 [Google Scholar](#)
34. Zacher B, Michel M, Schwalb B, Cramer P, Tresch A, Gagneur J (2017) **Accurate Promoter and Enhancer Identification in 127 ENCODE and Roadmap Epigenomics Cell Types and Tissues by GenoSTAN** *PLoS One* **12**:e0169249 [Google Scholar](#)
35. Bonnet J, Lindeboom RGH, Pokrovsky D, Stricker G, Çelik MH, Rupp RAW, et al. (2019) **Quantification of Proteins and Histone Marks in Drosophila Embryos Reveals Stoichiometric Relationships Impacting Chromatin Regulation** *Dev Cell* **51**:632–644 [Google Scholar](#)
36. Sneeringer CJ, Scott MP, Kuntz KW, Knutson SK, Pollock RM, Richon VM, et al. (2010) **Coordinated activities of wild-type plus mutant EZH2 drive tumor-associated hypertrimethylation of lysine 27 on histone H3 (H3K27) in human B-cell lymphomas** *Proceedings of the National Academy of Sciences* **107**:20980–5 [Google Scholar](#)
37. Zheng Y, Sweet SMM, Popovic R, Martinez-Garcia E, Tipton JD, Thomas PM, et al. (2012) **Total kinetic analysis reveals how combinatorial methylation patterns are established on lysines 27 and 36 of histone H3** *Proc Natl Acad Sci U S A* **109**:13549–54 [Google Scholar](#)
38. Lundkvist MJ, Lizana L, Schwartz YB (2023) **Forecasting histone methylation by Polycomb complexes with minute-scale precision** *Sci Adv* **9**:eadj8198 [Google Scholar](#)
39. Steffen PA, Fonseca JP, Gänger C, Dworschak E, Kockmann T, Beisel C, et al. (2013) **Quantitative in vivo analysis of chromatin binding of Polycomb and Trithorax group proteins reveals retention of ASH1 on mitotic chromatin** *Nucleic Acids Res* **41**:5235–50 [Google Scholar](#)
40. Nien CY, Liang HL, Butcher S, Sun Y, Fu S, Gocha T, et al. (2011) **Temporal coordination of gene networks by Zelda in the early Drosophila embryo** *PLoS Genet* **7**:e1002339 [Google Scholar](#)
41. Wang Z, Lindquist S (1998) **Developmentally regulated nuclear transport of transcription factors in Drosophila embryos enable the heat shock response** *Development* **125**:4841–

50 [Google Scholar](#)

42. Falahati H, Pelham-Webb B, Blythe S, Wieschaus E (2016) **Nucleation by rRNA Dictates the Precision of Nucleolus Assembly** *Curr Biol* **26**:277–85 [Google Scholar](#)
43. Rushlow CA, Han K, Manley JL, Levine M (1989) **The graded distribution of the dorsal morphogen is initiated by selective nuclear transport in Drosophila** *Cell* **59**:1165–77 [Google Scholar](#)
44. Gregor T, Wieschaus EF, McGregor AP, Bialek W, Tank DW (2007) **Stability and nuclear dynamics of the bicoid morphogen gradient** *Cell* **130**:141–52 [Google Scholar](#)
45. Gaskill MM, Soluri IV, Branks AE, Boka AP, Stadler MR, Vietor K, et al. (2023) **Localization of the Drosophila pioneer factor GAF to subnuclear foci is driven by DNA binding and required to silence satellite repeat expression** *Dev Cell* **58**:1610–1624 [Google Scholar](#)
46. Raff JW, Kellum R, Alberts B (1994) **The Drosophila GAGA transcription factor is associated with specific regions of heterochromatin throughout the cell cycle** *EMBO J* **13**:5977–83 [Google Scholar](#)
47. Breen TR, Duncan IM (1986) **Maternal expression of genes that regulate the bithorax complex of Drosophila melanogaster** *Developmental Biology* **118**:442–56 [Google Scholar](#)
48. Beatty RA, Waddington CH. XVIII (1949) —**Studies on Reproduction in Wild-Type and Female-Sterile Mutants of Drosophila melanogaster (Meig)**. *Proceedings of the Royal Society of Edinburgh Section B: Biological Sciences* **63**:249–70 [Google Scholar](#)
49. Bejarano F, Busturia A (2004) **Function of the Trithorax-like gene during Drosophila development** *Dev Biol* **268**:327–41 [Google Scholar](#)
50. Busturia A, Lloyd A, Bejarano F, Zavortink M, Xin H, Sakonju S (2001) **The MCP silencer of the Drosophila Abd-B gene requires both Pleiohomeotic and GAGA factor for the maintenance of repression** *Development* **128**:2163–73 [Google Scholar](#)
51. Americo J, Whiteley M, Brown JL, Fujioka M, Jaynes JB, Kassis JA (2002) **A complex array of DNA-binding proteins required for pairing-sensitive silencing by a polycomb group response element from the Drosophila engrailed gene** *Genetics* **160**:1561–71 [Google Scholar](#)
52. Mishra RK, Mihaly J, Barges S, Spierer A, Karch F, Hagstrom K, et al. (2001) **The iab-7 polycomb response element maps to a nucleosome-free region of chromatin and requires both GAGA and pleiohomeotic for silencing activity** *Mol Cell Biol* **21**:1311–8 [Google Scholar](#)
53. Soruco MML, Chery J, Bishop EP, Siggers T, Tolstorukov MY, Leydon AR, et al. (2013) **The CLAMP protein links the MSL complex to the X chromosome during Drosophila dosage compensation** *Genes Dev* **27**:1551–6 [Google Scholar](#)
54. Lehmann M, Siegmund T, Lintermann KG, Korge G (1998) **The pipsqueak protein of Drosophila melanogaster binds to GAGA sequences through a novel DNA-binding domain** *J Biol Chem* **273**:28504–9 [Google Scholar](#)
55. Tolhuis B, Muijters I, de Wit E, Teunissen H, Talhout W, van Steensel B, et al. (2006) **Genome-wide profiling of PRC1 and PRC2 Polycomb chromatin binding in Drosophila melanogaster** *Nat Genet* **38**:694–9 [Google Scholar](#)

56. Seller CA, Cho CY, O'Farrell PH (2019) **Rapid embryonic cell cycles defer the establishment of heterochromatin by Eggless/SetDB1 in *Drosophila*** *Genes Dev* **33**:403–17 [Google Scholar](#)
57. Love MI, Huber W, Anders S (2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biol* **15**:550 [Google Scholar](#)
58. Qian S, Capovilla M, Pirrotta V (1991) **The bx region enhancer, a distant cis-control element of the *Drosophila* Ubx gene and its regulation by hunchback and other segmentation genes** *EMBO J* **10**:1415–25 [Google Scholar](#)
59. Hannon CE, Blythe SA, Wieschaus EF (2017) **Concentration dependent chromatin states induced by the bicoid morphogen gradient** *eLife* **6**:e28275 <https://doi.org/10.7554/eLife.28275> | [Google Scholar](#)
60. Sun Y, Nien CY, Chen K, Liu HY, Johnston J, Zeitlinger J, et al. (2015) **Zelda overcomes the high intrinsic nucleosome barrier at enhancers during *Drosophila* zygotic genome activation** *Genome Res* **25**:1703–14 [Google Scholar](#)
61. Schulz KN, Bondra ER, Moshe A, Villalta JE, Lieb JD, Kaplan T, et al. (2015) **Zelda is differentially required for chromatin accessibility, transcription factor binding, and gene expression in the early *Drosophila* embryo** *Genome Res* **25**:1715–26 [Google Scholar](#)
62. McDaniel SL, Gibson TJ, Schulz KN, Fernandez Garcia M, Nevil M, Jain SU, et al. (2019) **Continued Activity of the Pioneer Factor Zelda Is Required to Drive Zygotic Genome Activation** *Mol Cell* **74**:185–195 [Google Scholar](#)
63. Hamm DC, Larson ED, Nevil M, Marshall KE, Bondra ER, Harrison MM (2017) **A conserved maternal- specific repressive domain in Zelda revealed by Cas9-mediated mutagenesis in *Drosophila melanogaster*** *PLoS Genet* **13**:e1007120 [Google Scholar](#)
64. Kvon EZ, Stampfel G, Yáñez-Cuna JO, Dickson BJ, Stark A (2012) **HOT regions function as patterned developmental enhancers and have a distinct cis-regulatory signature** *Genes Dev* **26**:908–13 [Google Scholar](#)
65. Stampfel G, Kazmar T, Frank O, Wienerroither S, Reiter F, Stark A (2015) **Transcriptional regulators form diverse groups with context-dependent regulatory functions** *Nature* :528 [Google Scholar](#)
66. Yamada S, Whitney PH, Huang SK, Eck EC, Garcia HG, Rushlow CA (2019) **The *Drosophila* Pioneer Factor Zelda Modulates the Nuclear Microenvironment of a Dorsal Target Enhancer to Potentiate Transcriptional Output** *Curr Biol* **29**:1387–1393 [Google Scholar](#)
67. Mir M, Stadler MR, Ortiz SA, Hannon CE, Harrison MM, Darzacq X, et al. (2018) **Dynamic multifactor hubs interact transiently with sites of active transcription in *Drosophila* embryos** *eLife* **7**:e40497 <https://doi.org/10.7554/eLife.40497> | [Google Scholar](#)
68. Dufourt J, Trullo A, Hunter J, Fernandez C, Lazaro J, Dejean M, et al. (2018) **Temporal control of gene expression by the pioneer factor Zelda through transient interactions in hubs** *Nat Commun* **9**:5194 [Google Scholar](#)
69. Hug CB, Grimaldi AG, Kruse K, Vaquerizas JM (2017) **Chromatin Architecture Emerges during Zygotic Genome Activation Independent of Transcription** *Cell* **169**:216–228 [Google Scholar](#)

70. Blythe SA, Wieschaus EF (2015) **Zygotic genome activation triggers the DNA replication checkpoint at the midblastula transition** *Cell* **160**:1169–81 [Google Scholar](#)
71. Ghotbi E, Ye P, Ervin T, Kum A, Benes J, Jones RS (2021) **Polycomb-group recruitment to a Drosophila target gene is the default state that is inhibited by a transcriptional activator** *Sci Adv* **7**:eabg1556 [Google Scholar](#)
72. Schuettengruber B, Ganapathi M, Leblanc B, Portoso M, Jaschek R, Tolhuis B, et al. (2009) **Functional Anatomy of Polycomb and Trithorax Chromatin Landscapes in Drosophila Embryos** *PLOS Biology* **7**:e1000013 [Google Scholar](#)
73. Schwartz YB, Kahn TG, Nix DA, Li XY, Bourgon R, Biggin M, et al. (2006) **Genome-wide analysis of Polycomb targets in Drosophila melanogaster** *Nat Genet* **38**:700–5 [Google Scholar](#)
74. Schaaf CA, Misulovin Z, Gause M, Koenig A, Gohara DW, Watson A, et al. (2013) **Cohesin and Polycomb Proteins Functionally Interact to Control Transcription at Silenced and Active Genes** *PLOS Genetics* **9**:e1003560 [Google Scholar](#)
75. Kang H, Jung YL, McElroy KA, Zee BM, Wallace HA, Woolnough JL, et al. (2017) **Bivalent complexes of PRC1 with orthologs of BRD4 and MOZ/MORF target developmental genes in Drosophila** *Genes Dev* **31**:1988–2002 [Google Scholar](#)
76. Pherson M, Misulovin Z, Gause M, Mihindukulasuriya K, Swain A, Dorsett D (2017) **Polycomb repressive complex 1 modifies transcription of active genes** *Sci Adv* **3**:e1700944 [Google Scholar](#)
77. Loubière V, Delest A, Thomas A, Bonev B, Schuettengruber B, Sati S, et al. (2016) **Coordinate redeployment of PRC1 proteins suppresses tumor formation during Drosophila development** *Nat Genet* **48**:1436–42 [Google Scholar](#)
78. Marsh AJ, Pirogov S, Kaur Y, Ruffridge AJ, Sajwan S, Gibson TJ, et al. (2025) **Catalytic-dependent and independent functions of the histone acetyltransferase CBP promote pioneer-factor-mediated zygotic genome activation** *Mol Cell* **85**:2409–2424 [Google Scholar](#)
79. Galouzis CC, Kherdjemil Y, Forneris M, Viales RR, Marco-Ferreres R, Furlong EEM (2025) **Chip (Ldb1) is a putative cofactor of Zelda forming a functional bridge to CBP during zygotic genome activation** *Mol Cell* **85**:2425–2441 [Google Scholar](#)
80. Mito Y, Henikoff JG, Henikoff S (2007) **Histone replacement marks the boundaries of cis-regulatory domains** *Science* **315** [Google Scholar](#)
81. Ogiyama Y, Schuettengruber B, Papadopoulos GL, Chang JM, Cavalli G (2018) **Polycomb-Dependent Chromatin Looping Contributes to Gene Silencing during Drosophila Development** *Mol Cell* **71**:73–88 [Google Scholar](#)
82. Loubiere V, Martinez AM, Cavalli G (2019) **Cell Fate and Developmental Regulation Dynamics by Polycomb Proteins and 3D Genome Architecture** *Bioessays* **41**:e1800222 [Google Scholar](#)
83. Larson ED, Komori H, Gibson TJ, Ostgaard CM, Hamm DC, Schnell JM, et al. (2021) **Cell-type-specific chromatin occupancy by the pioneer factor Zelda drives key developmental transitions in Drosophila** *Nat Commun* **12**:7153 [Google Scholar](#)
84. Staudt N, Fellert S, Chung HR, Jäckle H, Vorbrüggen G (2006) **Mutations of the Drosophila Zinc Finger- encoding Gene vielfältig Impair Mitotic Cell Divisions and Cause Improper**

Chromosome Segregation *Mol Biol Cell* **17**:2356–65 [Google Scholar](#)

85. Fitz-James MH, Sabarís G, Sarkies P, Bantignies F, Cavalli G (2025) **Interchromosomal contacts between regulatory regions trigger stable transgenerational epigenetic inheritance in *Drosophila*** *Molecular Cell* **85**:677–691 [Google Scholar](#)
86. Mei H, Kozuka C, Hayashi R, Kumon M, Koseki H, Inoue A (2021) **H2AK119ub1 guides maternal inheritance and zygotic deposition of H3K27me3 in mouse embryos** *Nat Genet* **53**:539–50 [Google Scholar](#)
87. Tabuchi TM, Rechtsteiner A, Jeffers TE, Egelhofer TA, Murphy CT, Strome S (2018) ***Caenorhabditis elegans* sperm carry a histone-based epigenetic memory of both spermatogenesis and oogenesis** *Nat Commun* **9**:4310 [Google Scholar](#)
88. Ciabrelli F, Comoglio F, Fellous S, Bonev B, Ninova M, Szabo Q, et al. (2017) **Stable Polycomb-dependent transgenerational inheritance of chromatin states in *Drosophila*** *Nat Genet* **49**:876–86 [Google Scholar](#)
89. Matsuwaka M, Kumon M, Inoue A (2025) **H3K27 dimethylation dynamics reveal stepwise establishment of facultative heterochromatin in early mouse embryos** *Nat Cell Biol* **27**:28–38 [Google Scholar](#)
90. Iovino N, Ciabrelli F, Cavalli G (2013) **PRC2 controls *Drosophila* oocyte cell fate by repressing cell cycle genes** *Dev Cell* **26**:431–9 [Google Scholar](#)
91. Cline TW (1978) **Two closely linked mutations in *Drosophila melanogaster* that are lethal to opposite sexes and interact with daughterless** *Genetics* **90**:683–97 [Google Scholar](#)
92. Soluri IV, Zumerling LM, Payan Parra OA, Clark EG, Blythe SA (2020) **Zygotic pioneer factor activity of Odd-paired/Zic is necessary for late function of the *Drosophila* segmentation network** *eLife* **9**:e53916 <https://doi.org/10.7554/eLife.53916> | [Google Scholar](#)
93. Stern DL, Kim E, Behrman EL (2024) **The *Janelia Atalanta* plasmids provide a simple and efficient CRISPR/Cas9-mediated homology directed repair platform for *Drosophila*** *bioRxiv* <https://doi.org/10.1101/2023.06.17.545412> | [Google Scholar](#)
94. Stern DL (2017) **Tagmentation-Based Mapping (TagMap) of Mobile DNA Genomic Insertion Sites** *bioRxiv* <https://doi.org/10.1101/037762> | [Google Scholar](#)
95. Kirchhofer A, Helma J, Schmidthals K, Frauer C, Cui S, Karcher A, et al. (2010) **Modulation of protein properties in living cells using nanobodies** *Nat Struct Mol Biol* **17**:133–8 [Google Scholar](#)
96. Jones RS, Gelbart WM (1990) **Genetic analysis of the enhancer of zeste locus and its role in gene regulation in *Drosophila melanogaster*** *Genetics* **126**:185–99 [Google Scholar](#)
97. Martin M (2011) **Cutadapt removes adapter sequences from high-throughput sequencing reads** *EMBnet.journal* **17**:10–2 [Google Scholar](#)
98. Langmead B, Salzberg SL (2012) **Fast gapped-read alignment with Bowtie 2** *Nat Methods* **9**:357–9 [Google Scholar](#)
99. Lawrence M, Huber W, Pagès H, Aboyoun P, Carlson M, Gentleman R, et al. (2013) **Software for Computing and Annotating Genomic Ranges** *PLOS Computational Biology* **9**:e1003118 [Google](#)

Scholar

100. Morgan M, Pagès H, Obenchain V, Hayden N. (2020) **Rsamtools: Binary alignment (BAM), FASTA, variant call (BCF), and tabix file import** *Bioconductor* <https://bioconductor.org/packages/Rsamtools>
101. Zhang Y, Liu T, Meyer CA, Eeckhoutte J, Johnson DS, Bernstein BE, et al. (2008) **Model-based Analysis of ChIP-Seq (MACS)** *Genome Biology* **9**:R137 [Google Scholar](#)
102. Lawrence M, Gentleman R (2009) **Carey V. rtracklayer: an R package for interfacing with genome browsers** *Bioinformatics* **25**:1841–2 [Google Scholar](#)
103. Zacher B, Ertl J, Campos-Martin R, Gagneur J, Tresch A. (2020) **STAN: The Genomic State ANnotation Package**
104. Yoon S, Baik B, Park T, Nam D (2021) **Powerful p-value combination methods to detect incomplete association** *Sci Rep* **11**:6980 [Google Scholar](#)
105. Shah RN, Grzybowski AT, Cornett EM, Johnstone AL, Dickson BM, Boone BA, et al. (2018) **Examining the Roles of H3K4 Methylation States with Systematically Characterized Antibodies** *Mol Cell* **72**:162–177 [Google Scholar](#)

Author information

Natalie Gonzaga-Saavedra

Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, United States, Department of Molecular Biosciences, Northwestern University, Evanston, United States

ORCID iD: [0009-0003-1962-8308](https://orcid.org/0009-0003-1962-8308)

Eleanor A Degen

Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, United States, Department of Molecular Biosciences, Northwestern University, Evanston, United States

ORCID iD: [0000-0002-0832-4507](https://orcid.org/0000-0002-0832-4507)

Isabella V Soluri

Department of Molecular Biosciences, Northwestern University, Evanston, United States

ORCID iD: [0009-0002-8791-7251](https://orcid.org/0009-0002-8791-7251)

Corinne Croslyn

Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, United States, Department of Molecular Biosciences, Northwestern University, Evanston, United States

ORCID iD: [0000-0002-9190-7390](https://orcid.org/0000-0002-9190-7390)

Shelby A Blythe

Department of Molecular Biosciences, Northwestern University, Evanston, United States, National Institute for Theory and Mathematics in Biology, Northwestern University and the University of Chicago, Chicago, United States

For correspondence: shelby.blythe@northwestern.edu

Editors

Reviewing Editor

Xin Chen

Johns Hopkins University, Baltimore, United States of America

Senior Editor

Adèle Marston

University of Edinburgh, Edinburgh, United Kingdom

Reviewer #1 (Public review):

This well-conceived manuscript investigates the mechanisms that shape the chromatin landscape following fertilization, using the *Drosophila* embryo as a model system. Importantly, the authors revisit conflicting data using new approaches and analysis to show that the silent H3K27me3 mark deposited by PRC2 is established de novo in the embryo in coordination with the slowing of the nuclear division cycle and activation of zygotic transcription. Unexpectedly, they demonstrate that the transcription factor GAF is not required for the deposition of this mark, but that the well-studied pioneer factor Zelda, which is required for widespread gene expression, is required for H3K27me3 deposition at a subset of regions. The experiments are rigorously performed, and interpretations are clear. Strengths of this manuscript include the rigor of the experimental design, careful analysis, and well-supported conclusions. Some additional citations, analysis, and broadening of the Discussion section to include additional models and data would further strengthen this manuscript.

<https://doi.org/10.7554/eLife.108371.1.sa8>

Reviewer #2 (Public review):

Epigenetic silencing of target genes by the Polycomb pathway is central to maintenance of cell fates during development and depends on repressive chromatin states involving Polycomb complexes and histone modifications. However, the mechanisms by which these chromatin states are built at the earliest stages of development are unclear. Here, Gonzaga-Saavedra and colleagues use the premier experimental system for studying Polycomb gene regulation, *Drosophila* development, to investigate when Polycomb domains emerge and how they are assembled. Using a combination of CRISPR gene editing, imaging, and genomic profiling, they determine that while H3K27me3 is initially present in the first nuclear cycles, it quickly dissipates and does not re-emerge until mid-nuclear cycle 14, during the major wave of zygotic genome activation (ZGA). This finding helps resolve current discrepancies in the field, informs potential mechanisms of transgenerational inheritance, and indicates that repressive Polycomb domains are built de novo on target genes in embryogenesis. The authors then set out to examine how Polycomb domains are built. Through live imaging and immunofluorescence, they determine that the histone H3K27 methyltransferase, E(z), is present in nuclei at high levels throughout cleavage and blastoderm stages. By contrast, they determine that several Polycomb proteins that bind PREs (cis elements that demarcate Polycomb targets in the genome) are absent from early cleavage nuclei and progressively increase following nuclear cycle 10. These findings suggest that the absence of H3K27me3 in early embryos may be due to failure to assemble functional Polycomb complexes at target

genes. Lastly, the authors test the requirement of two transcription factors with important roles in ZGA, GAF, and ZLD. Despite binding to many PREs and regulating chromatin accessibility in early embryos, they find that GAF is largely dispensable for the emergence of H3K27me3 domains. On the other hand, they find that the pioneer factor ZLD is required for proper H3K27me3 emergence; in its absence, some Polycomb domains accumulate greater levels of H3K27me3, whereas other Polycomb domains accumulate less H3K27me3.

Strengths:

The strengths of this study are manifold. It studies an important topic with broad interest to the chromatin and epigenetics fields. It is well-written with detailed method descriptions. In addition, the experimental design and rigor of execution are exceptional despite working with very small amounts of biological material. Example strengths include that the Polycomb proteins studied were tagged with the same epitope, permitting direct quantitative comparisons in imaging and in genomics experiments. Microscopy studies are quantified and performed both via live imaging and via immunofluorescence. The microscopy studies reinforce and extend conclusions made via ChIP. Sophisticated loss-of-function analyses allow for direct mechanistic tests of Polycomb domain emergence.

Weaknesses:

Overall, the study is quite strong already, but it can be further strengthened in several ways. First, several conclusions should be refined based on the data presented. Second, the extent to which ZLD is important for initiating Polycomb domain formation should be made clearer. Third, additional genomic profiling experiments are needed to provide insight into models explaining why H3K27me3 is absent prior to NC14.

<https://doi.org/10.7554/eLife.108371.1.sa7>

Reviewer #3 (Public review):

Gonzaga-Saavedra et al report an analysis on genomic binding of Polycomb group proteins, and of H2Aub1 and H3K27me3 domain formation in the early *Drosophila* embryo. Using carefully staged embryos during the nuclear cycles (NC) leading up to the cellular blastoderm stage, the authors provide compelling evidence that H3K27me3 domains at PcG target genes are only established during NC14 and do not exist in NC13. In contrast, H2Aub1 domains already start to appear during NC13. The authors show that E(z), the catalytic subunit of the H3K27 histone methyltransferase PRC2, is readily detected in interphase nuclei during the rapid nuclear divisions in pre-blastoderm embryos. In contrast, the DNA-binding proteins Pho, Cg, and GAF that are known (Pho) or have been postulated (Cg, GAF) to anchor PRC2 and PRC1 to Polycomb Response Elements (PREs) in Polycomb target genes only start to show nuclear localization from NC10 onwards with gradually increasing nuclear concentrations, reaching a maximum during NC14. These data strongly corroborate the simple, straightforward view that targeting of PRC2 and PRC1 to PREs by sequence-specific DNA-binding proteins is a prerequisite for the formation of H3K27me3 and H2Aub1 domains at Polycomb target genes.

The authors then explore the potential role of GAF/Trl in this process. They find that in embryos depleted of GAF/Trl, H3K27me3 domain formation is largely unperturbed.

The authors also depleted the pioneer factor Zelda (Zld) and found that removal of Zld results in a more complex outcome. Zelda appears to counteract the accumulation of H3K27me3 at the Polycomb targets *eve* and *zen*, but also appears to be required for effective H3K27me3 domain formation at Polycomb targets such as *amos* or *atonal*.

This is a very thorough study that reports data of superior technical quality that are highly relevant for the field. The study by Gonzaga-Saavedra et al extends and strengthens previous work from the labs of Eisen (Li et al, eLife 2014) and Zeitlinger (Chen et al, eLife 2013) to convincingly demonstrate that Polycomb domain formation in the early embryo occurs during ZGA but that such domains do not exist prior to ZGA. This should now finally put to rest earlier claims by the Iovino lab (Zenk et al, Science 2017) that H3K27me3 domains present in the zygote nucleus would be propagated and partially maintained during the rapid nuclear cleavage cycles and serve as seeds for H3K27me3 domain formation during ZGA.

The experiments analyzing H3K27me3 domain formation in embryos depleted of GAF/Trl or Zelda will be of great interest to the field.

<https://doi.org/10.7554/eLife.108371.1.sa6>

Author response:

Reviewer 1:

We appreciate the reviewer's positive assessment and in revision will expand the Discussion to clarify some of the mechanistic insights of this work, as well as to include expanded treatment of related studies in other model systems.

Reviewer 2:

We are grateful for the reviewer's thorough and supportive comments. We will carefully revise assertions and conclusions for objectivity. Additional analysis of the Zelda experiments will be performed and experimental data tables will be updated to report these results. For the point about providing "insight into models explaining why H3K27me3 is absent prior to NC14," we have recently submitted a related preprint that addresses this issue directly (Degen, Gonzaga-Saavedra, and Blythe, bioRxiv 2025). In summary, we find evidence that a maternal PcG imprint is indeed maintained through cleavage divisions, albeit through lower-order methylation states (maximally H3K27me2). We chose not to include these additional results in this manuscript to maintain the focus of this study on ZGA. Our revision of this manuscript will include a section in the Discussion that synthesizes the conclusions of the two studies.

Reviewer 3:

We thank the reviewer for recognizing the strength of our data and conclusions, and we agree that our results help settle conflicting claims in the field. We will emphasize Zelda's context-dependent effects more clearly in the revised manuscript.

References:

Degen EA, Gonzaga-Saavedra N, Blythe SA. Lower-order methylation states underlie the maintenance and re-establishment of Polycomb modifications in *Drosophila* embryogenesis. bioRxiv [Preprint]. 2025 Jul 29:2025.07.25.666882. doi: 10.1101/2025.07.25.666882. PMID: 40766521; PMCID: PMC12324246.

<https://doi.org/10.7554/eLife.108371.1.sa5>